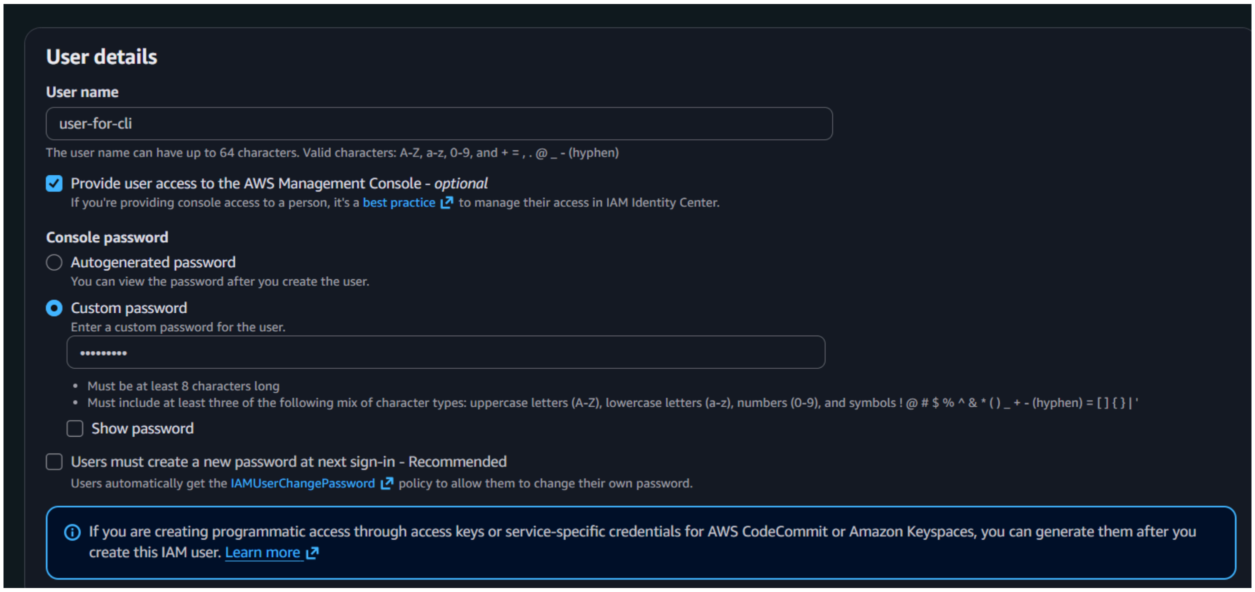
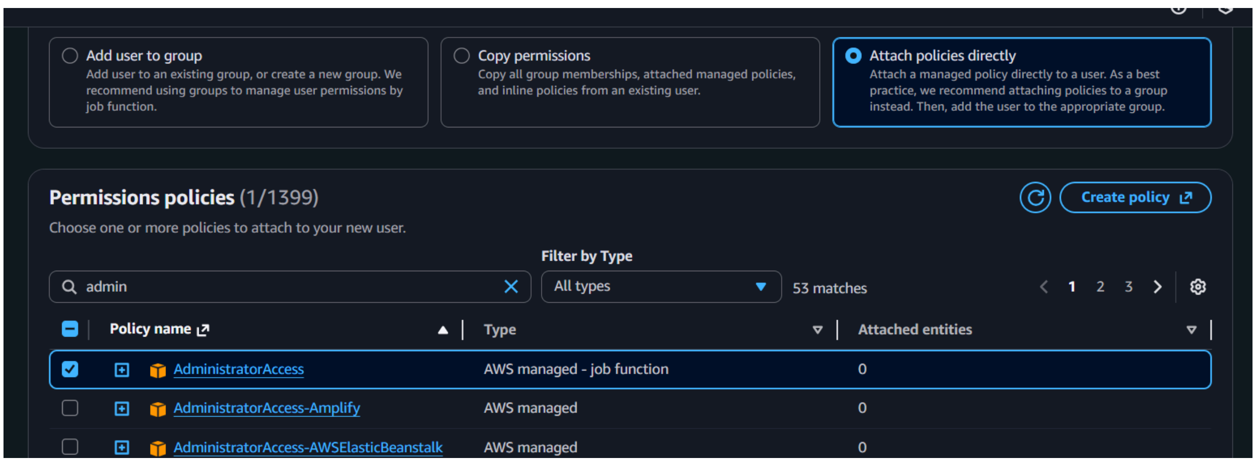
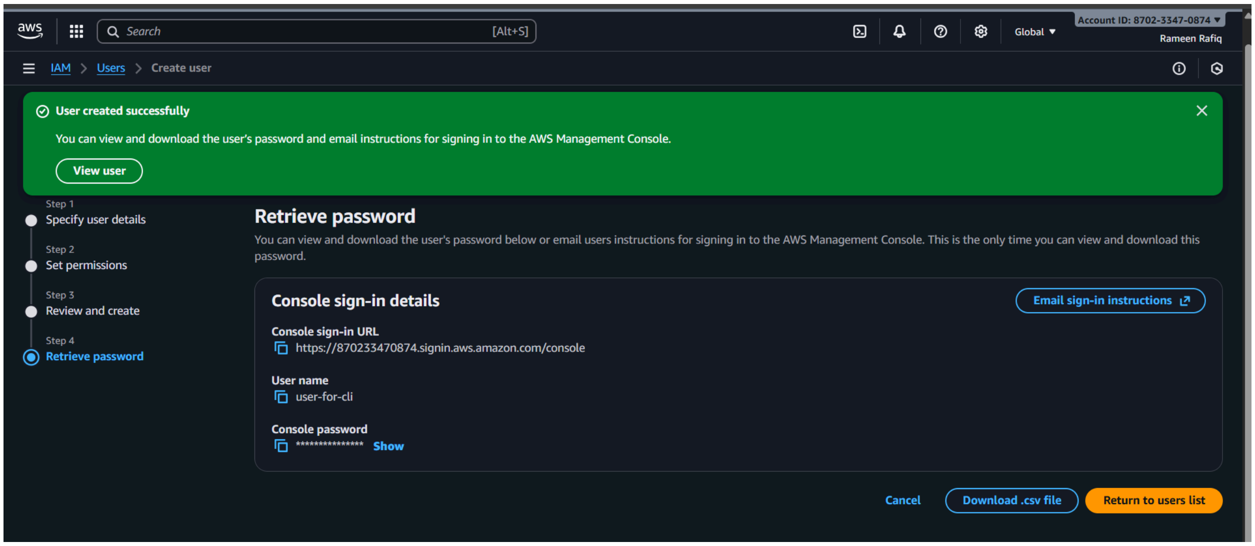
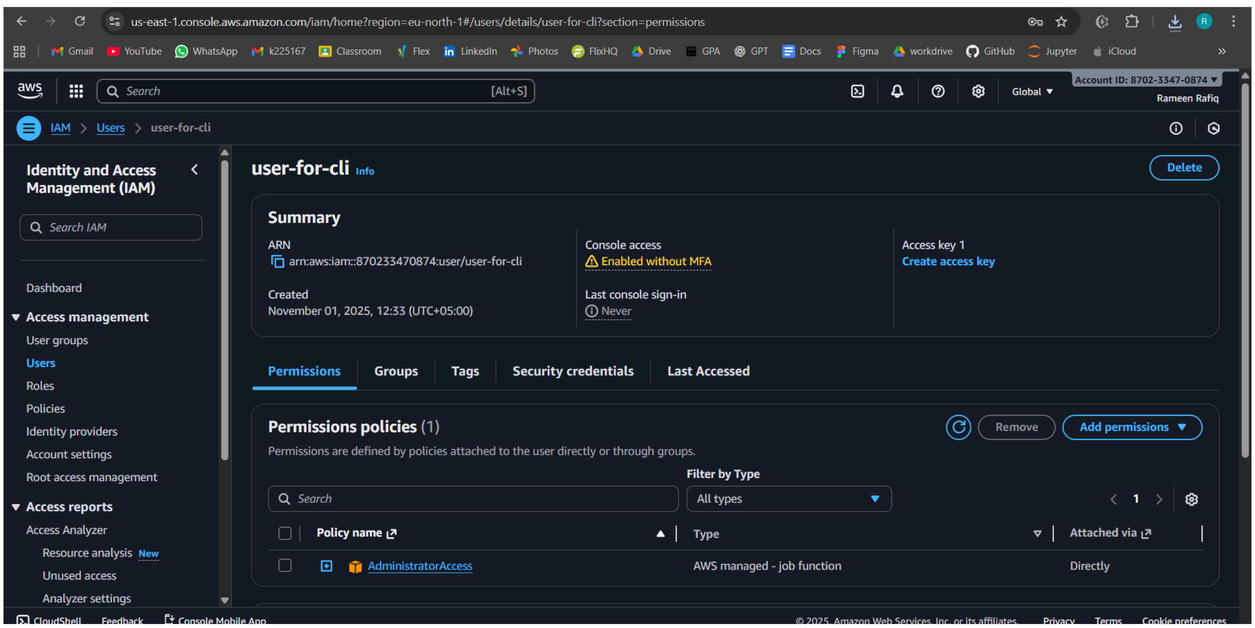
Log in as root user and make a new user account. We now access user through command line.

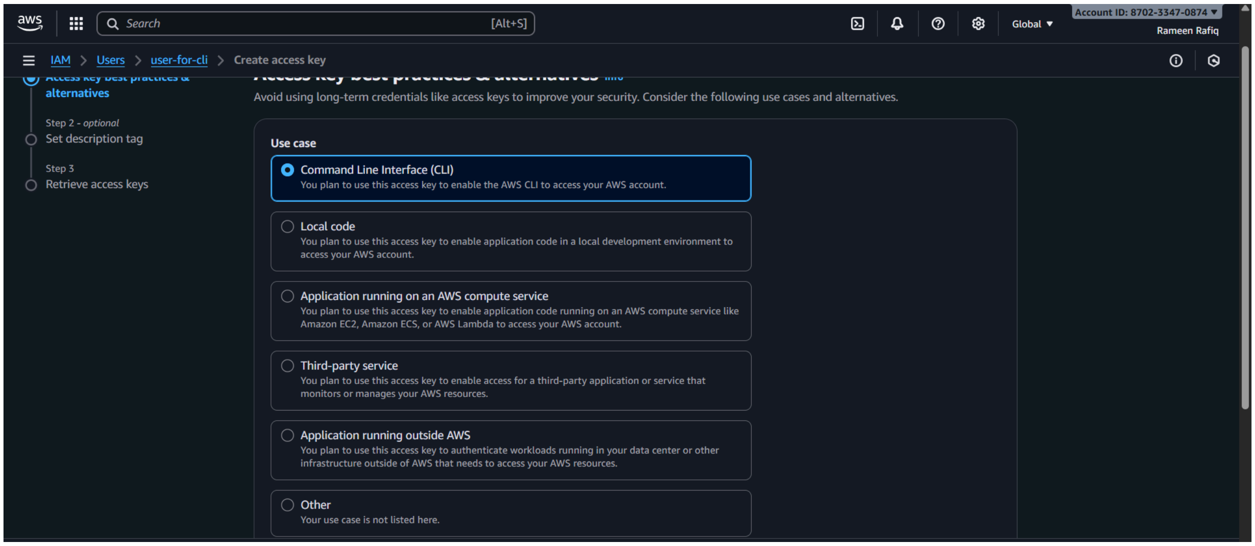


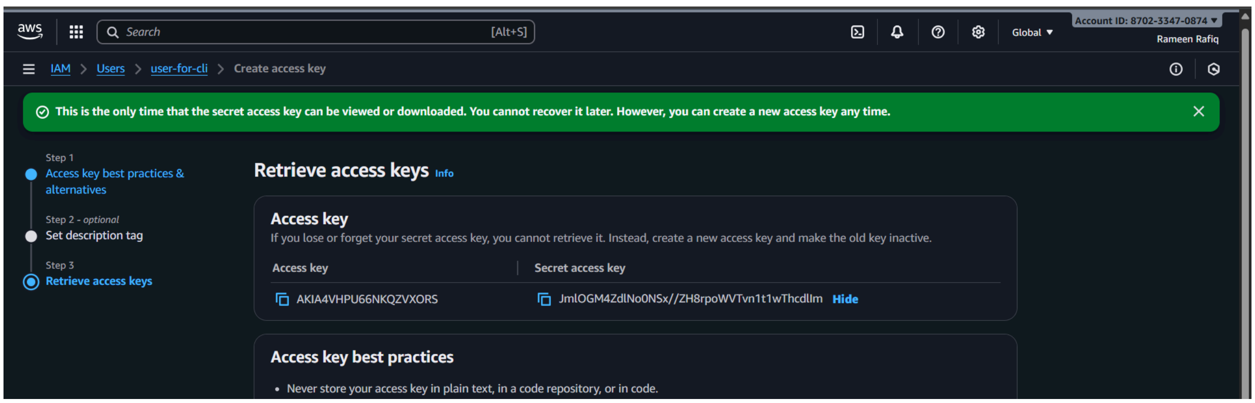






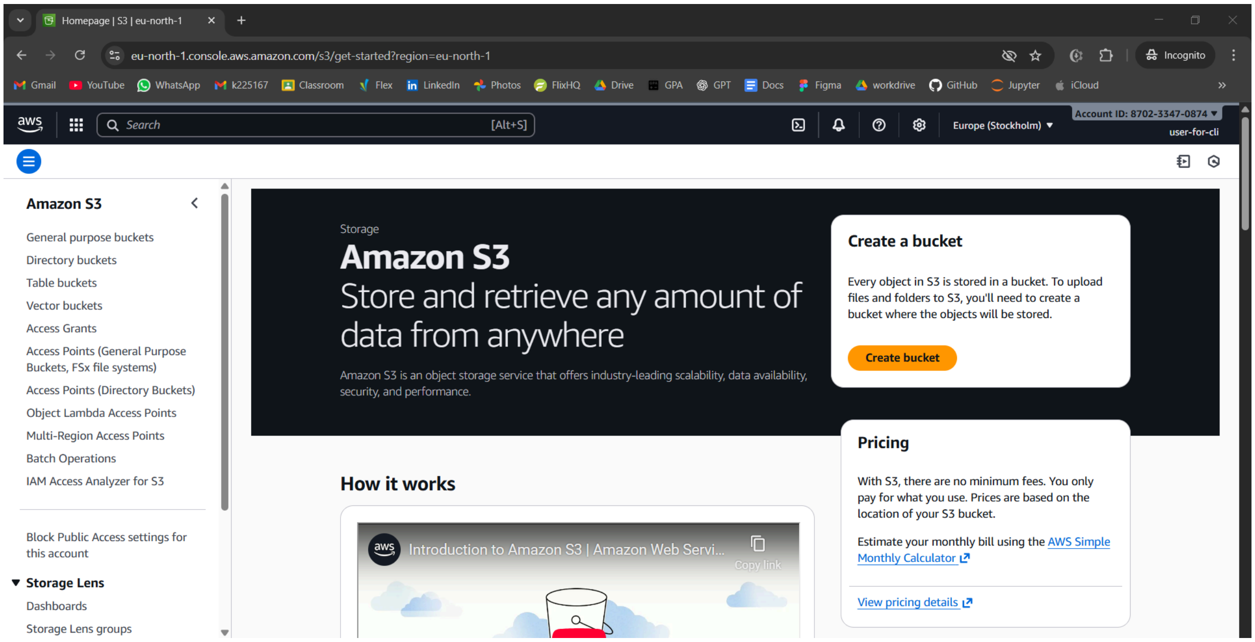
Note the create access key option



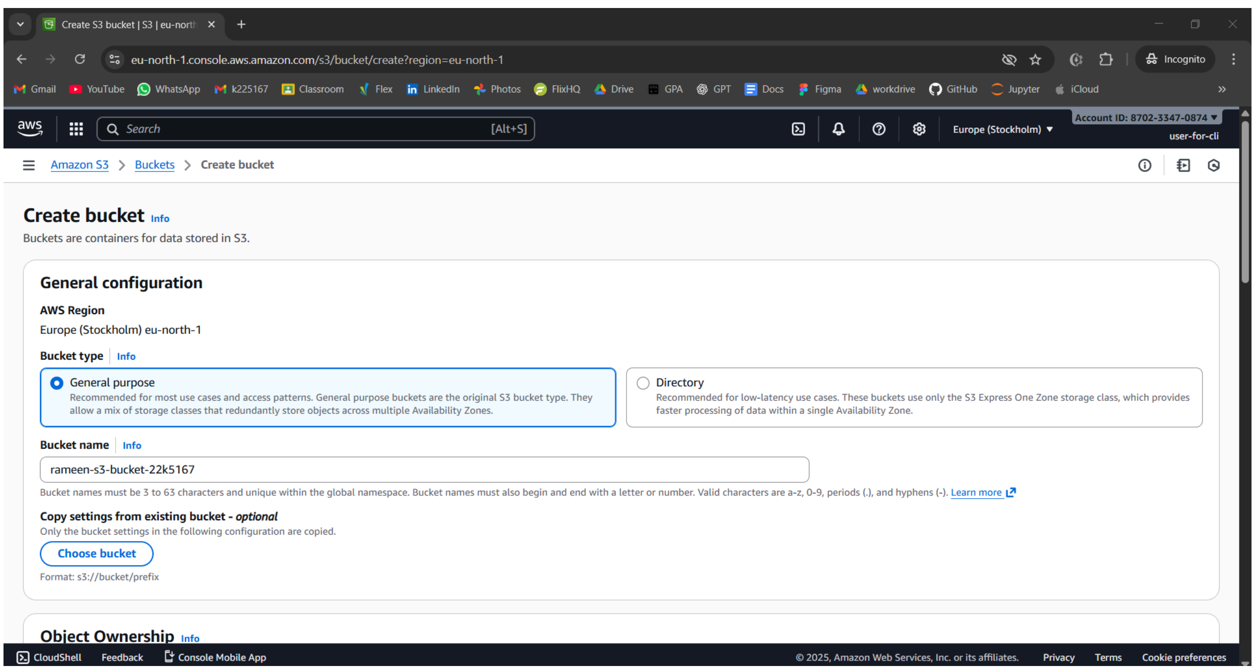


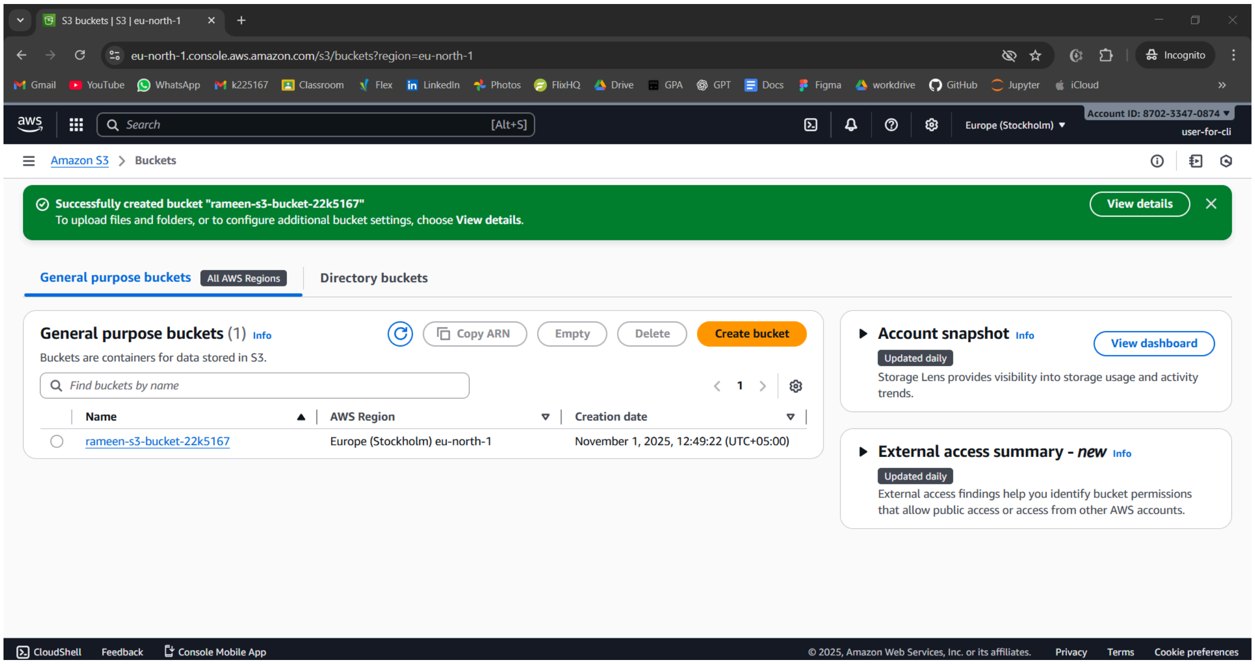
Access key: AKIA4VHPU66NKQZVXORS
Secret access key: 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

Now go to incognito and access the user just created
Go to s3

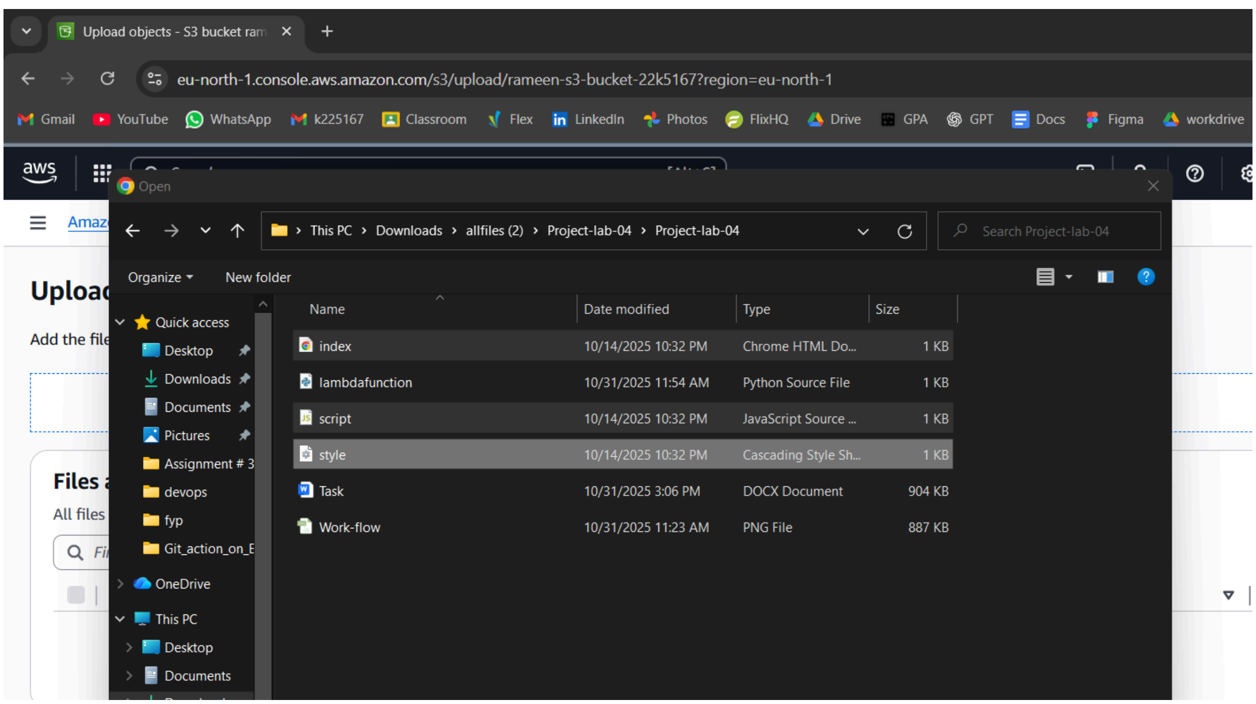


Now create a bucket

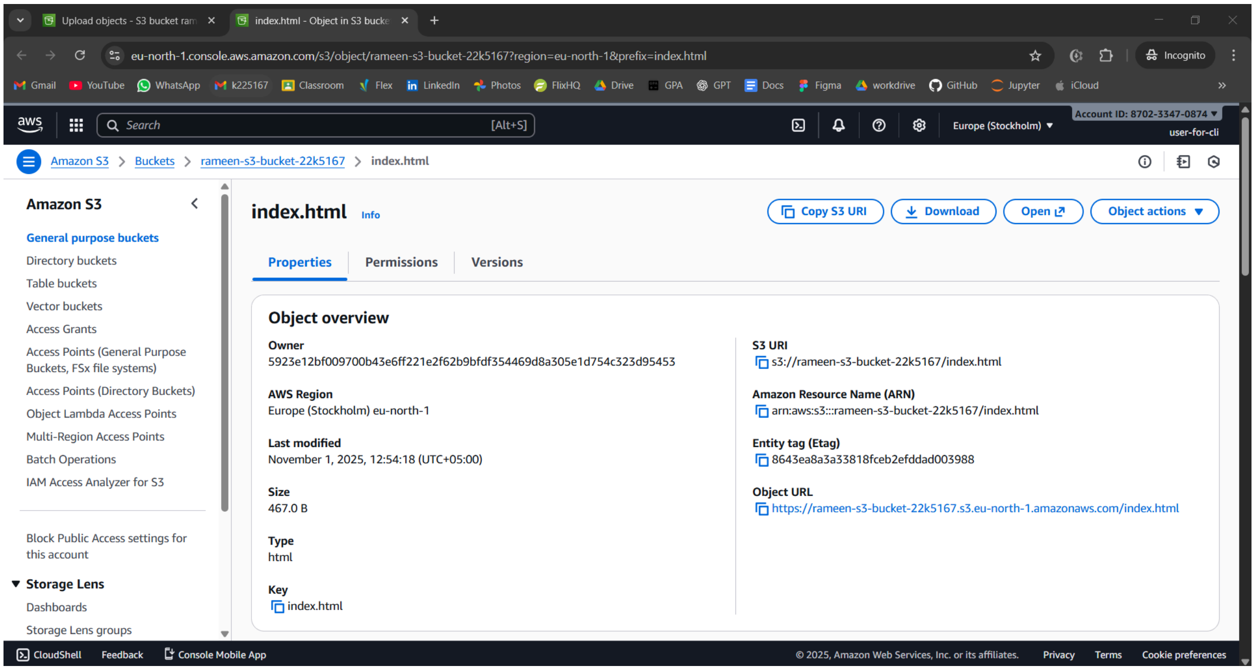




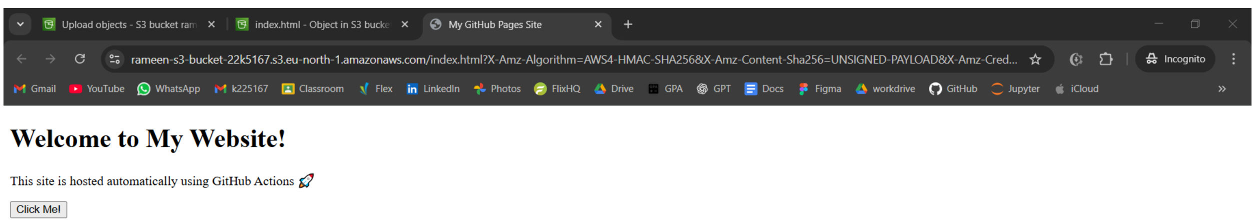
Upload files in bucket

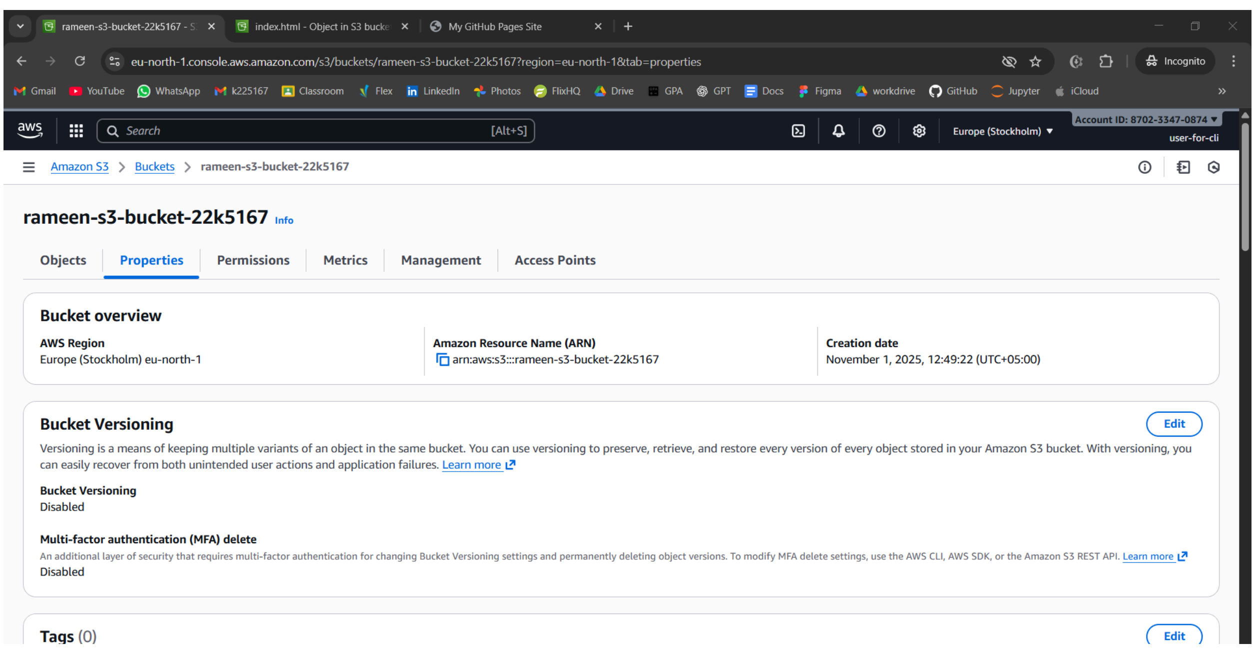


Options for a file

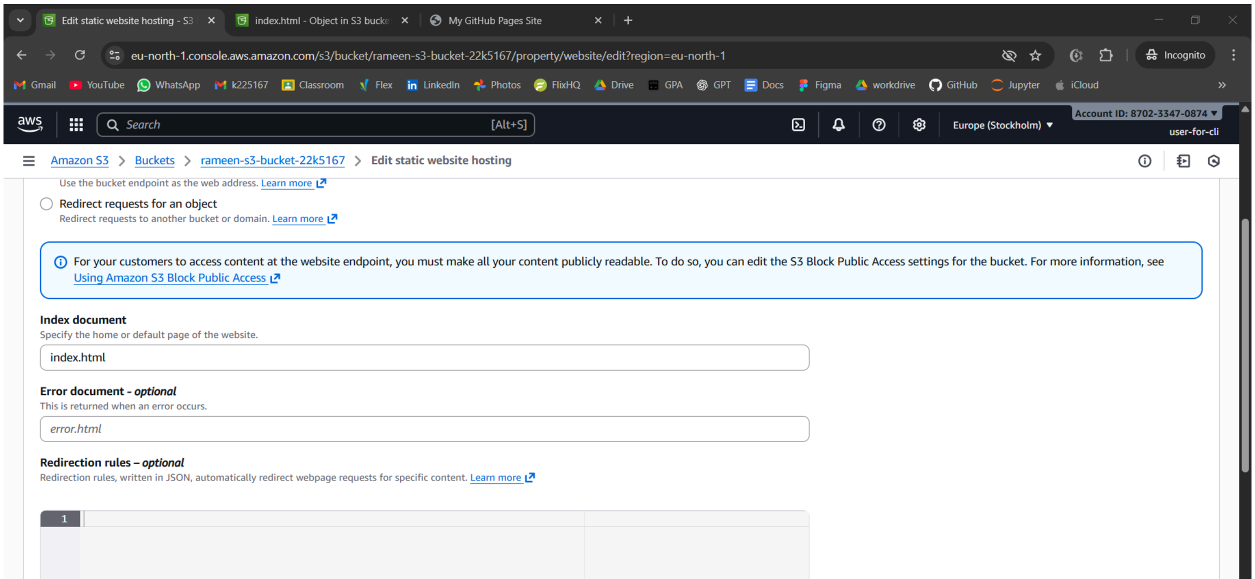


After opening the index file

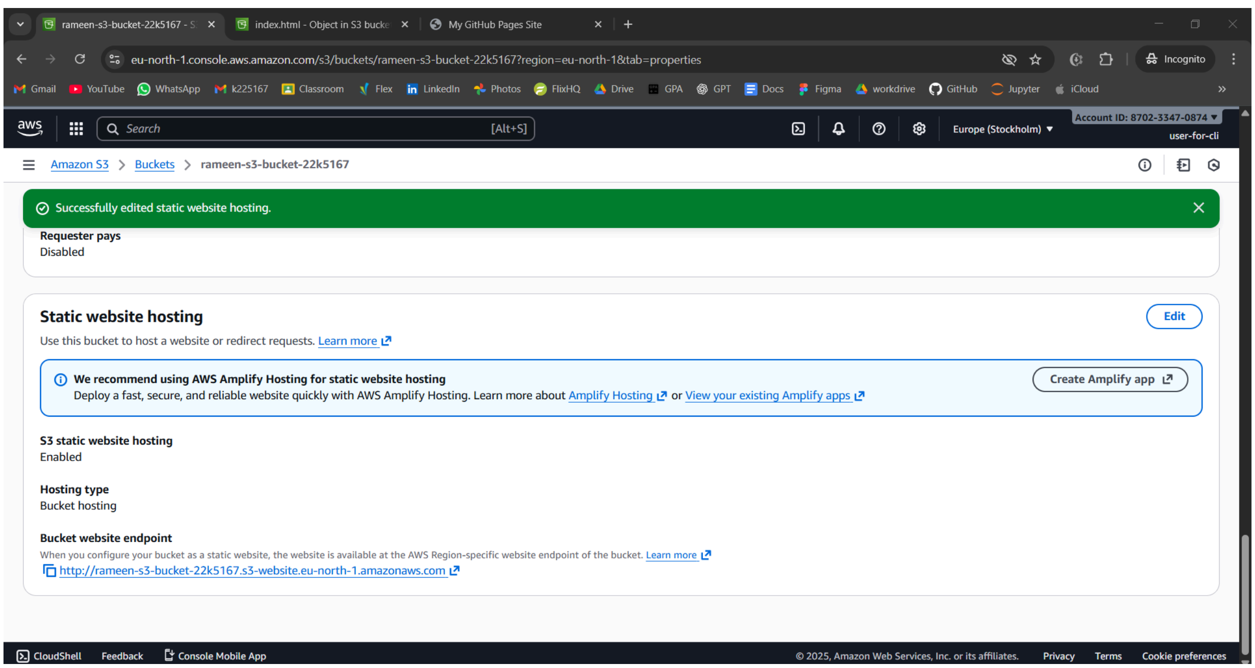




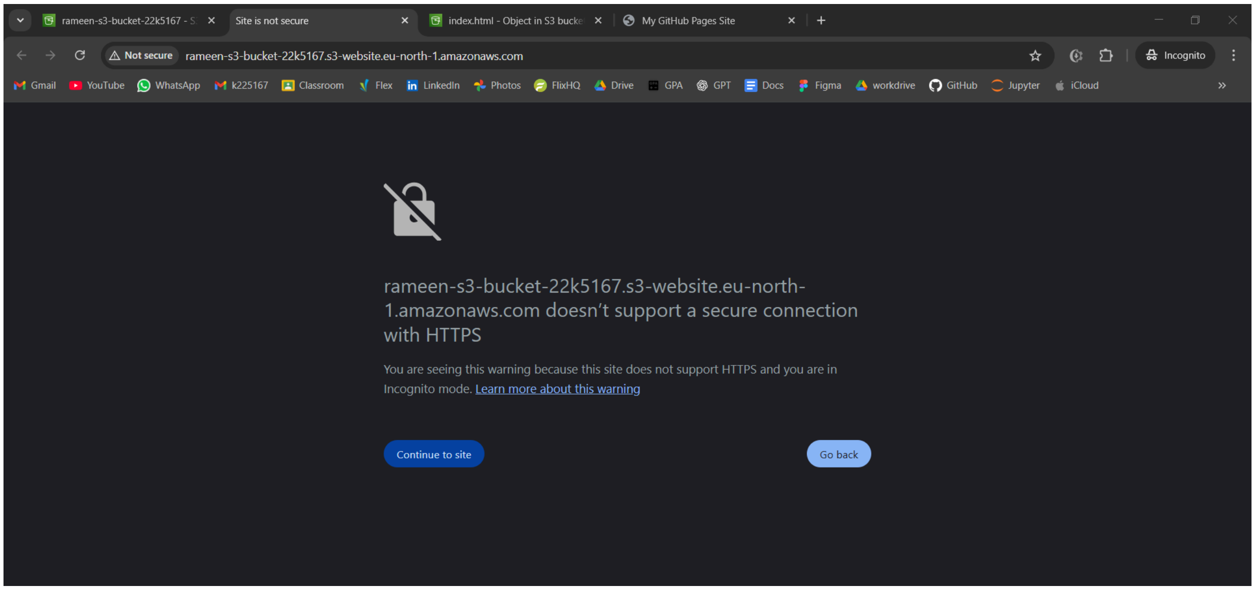
Come to end of page
Click on edit



Url being shown at the end now

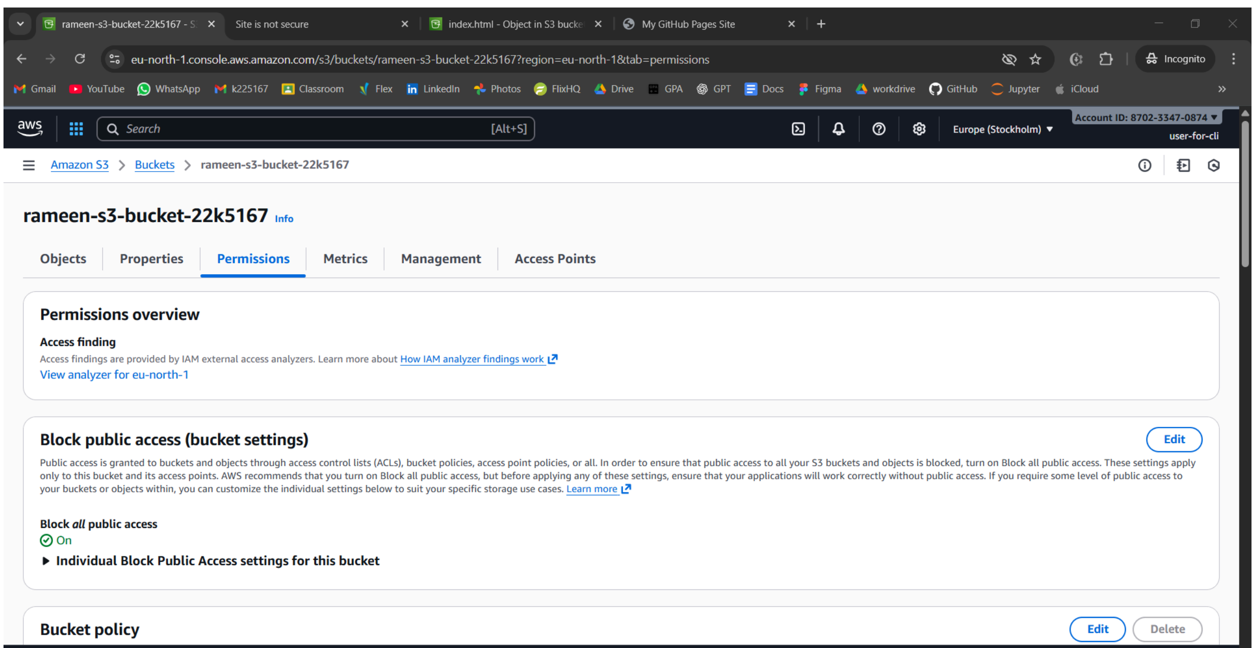


This shows error

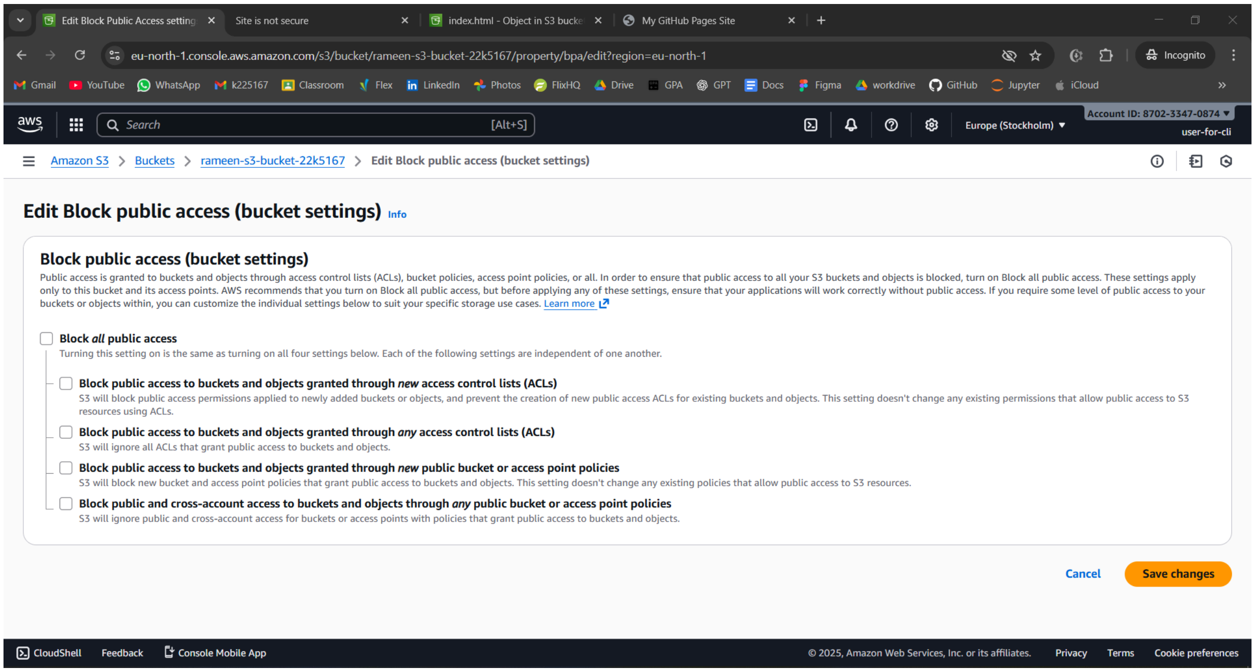


First we were just seeing the bucket. But then we are accessing the public url. So we need to give access

Now come here



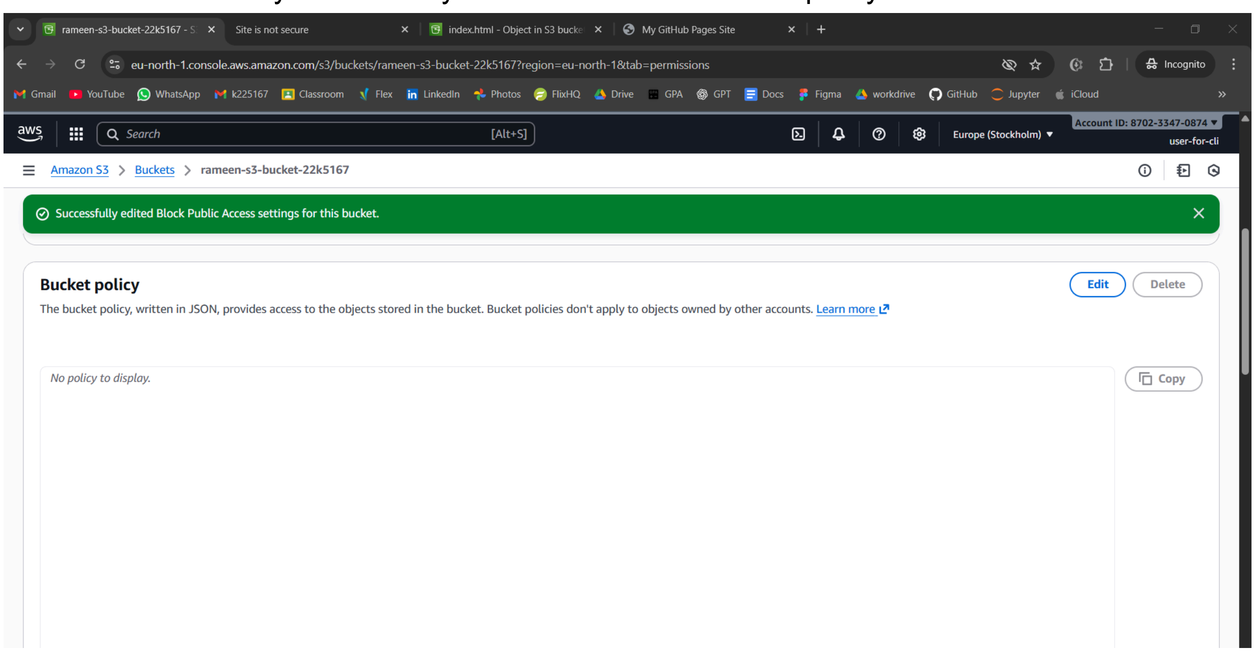
Turn off block

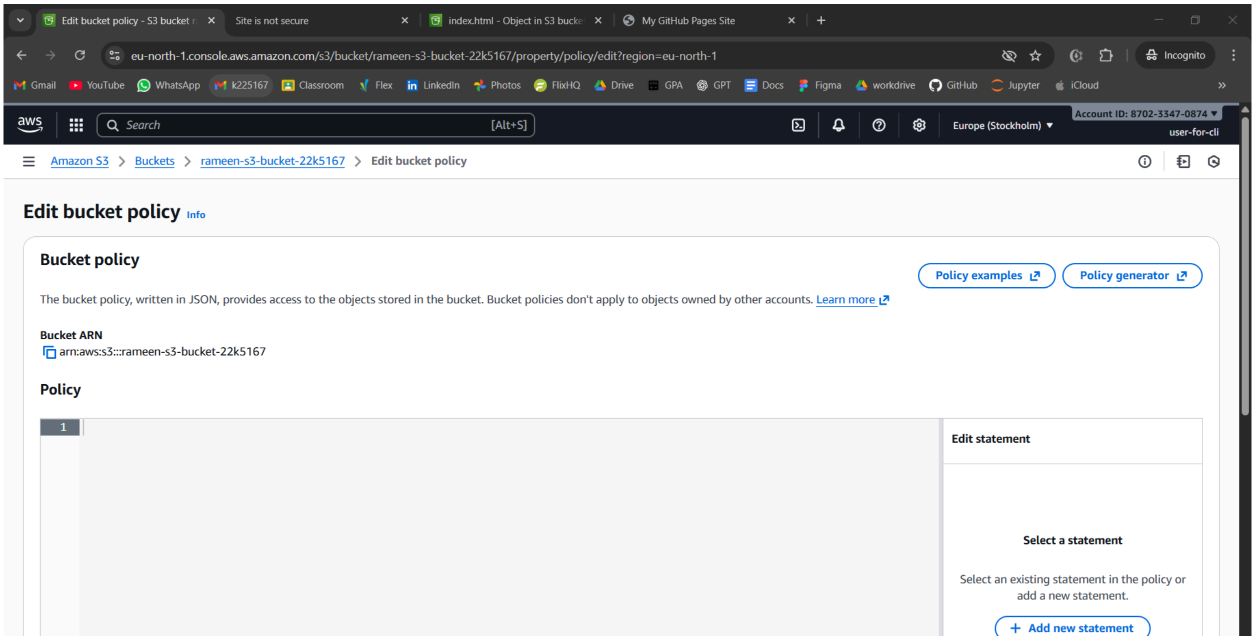


It still wont work, bec we just enabled to get permission, but not granted the permission. So grant permission now

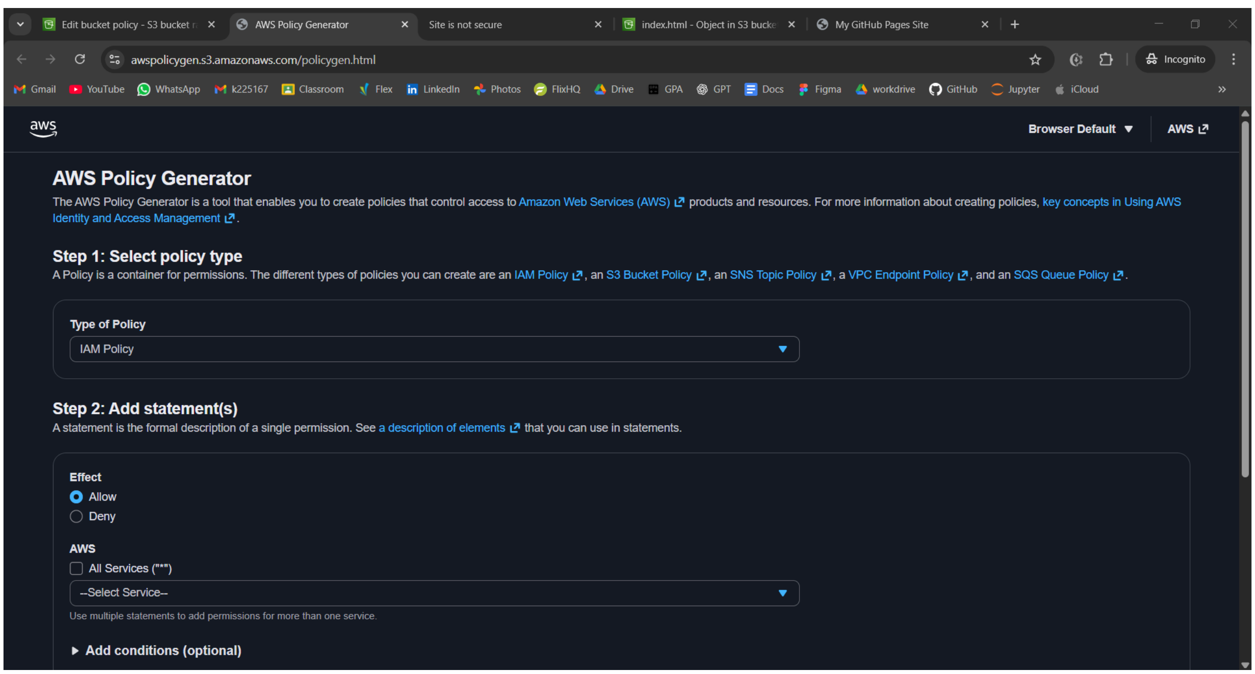
We need to grant permission here:

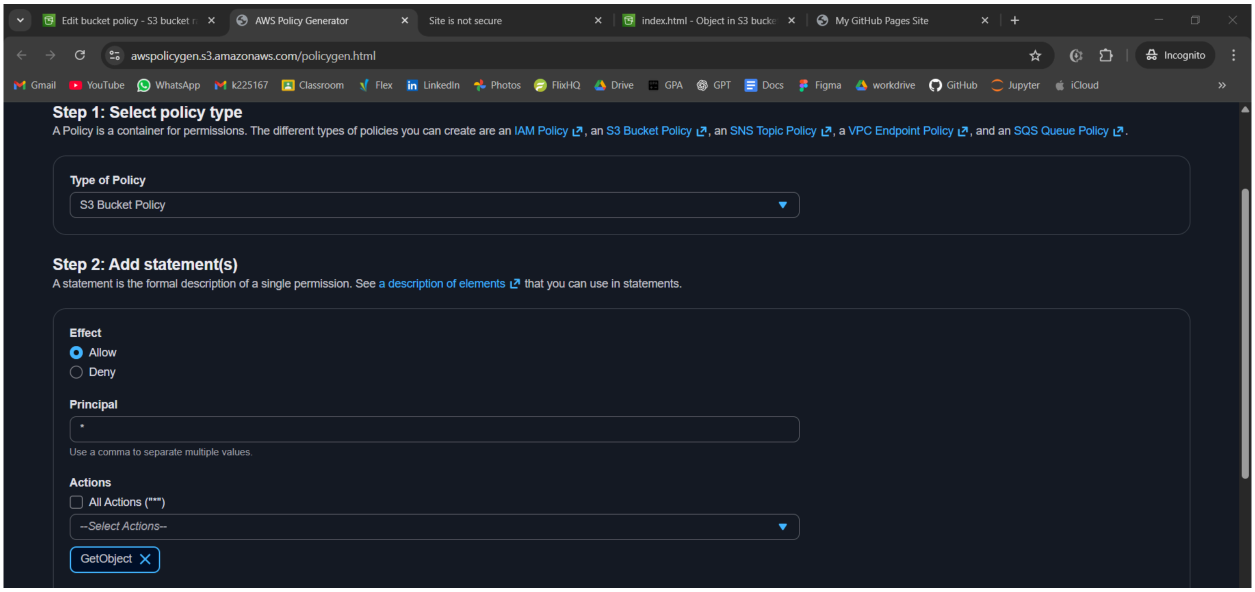
it acts like an extra layer of security. For this bucket we create policy ourself.



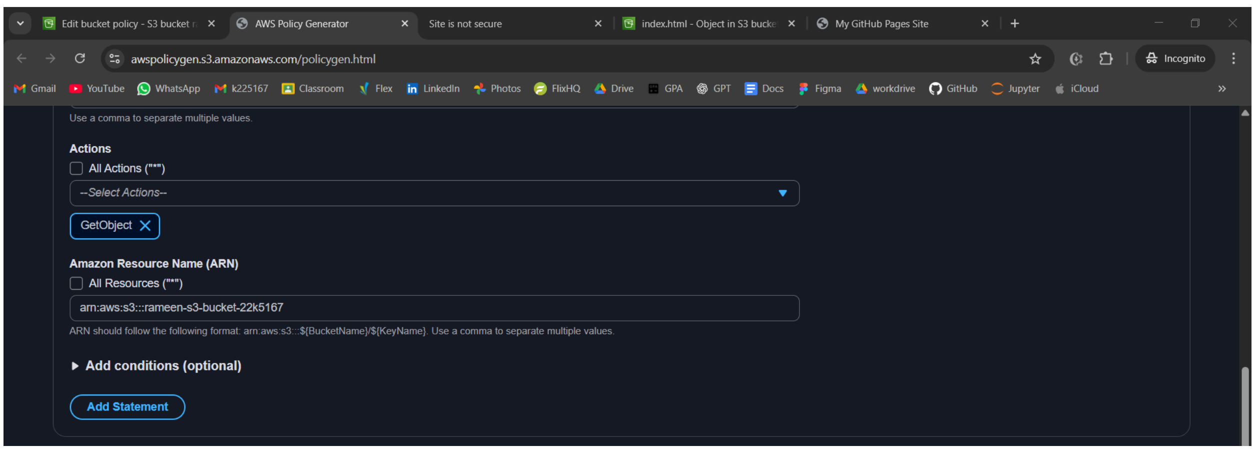


This interface will come:

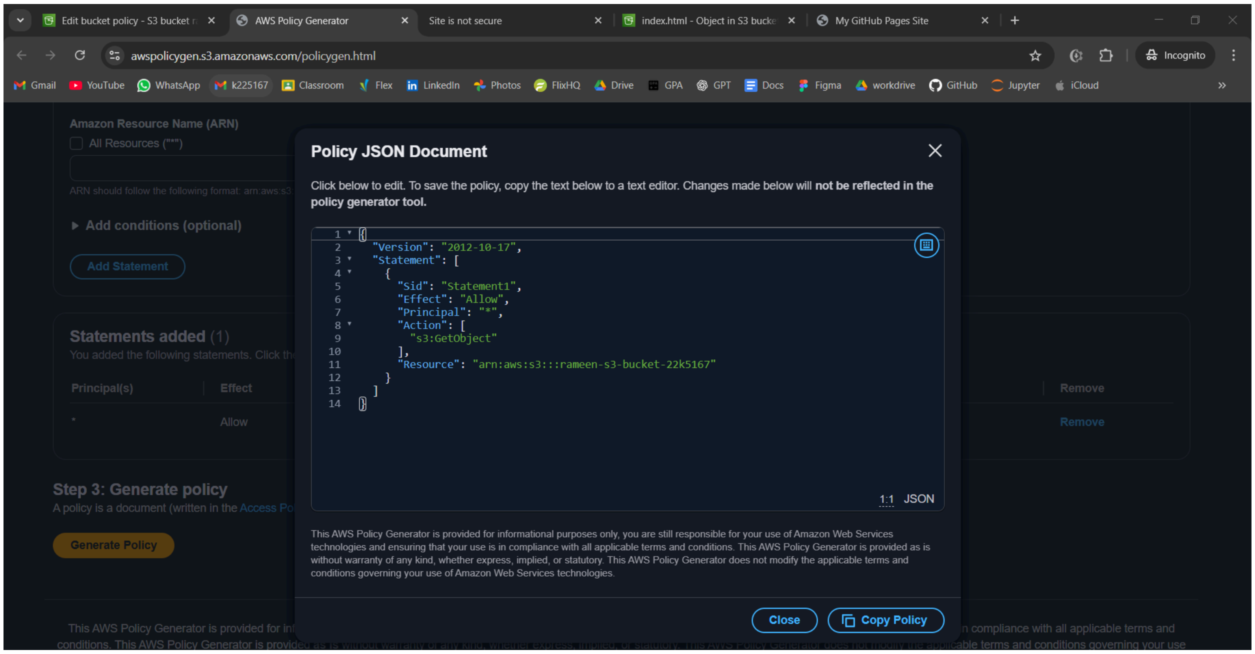




Copy the arn for bucket

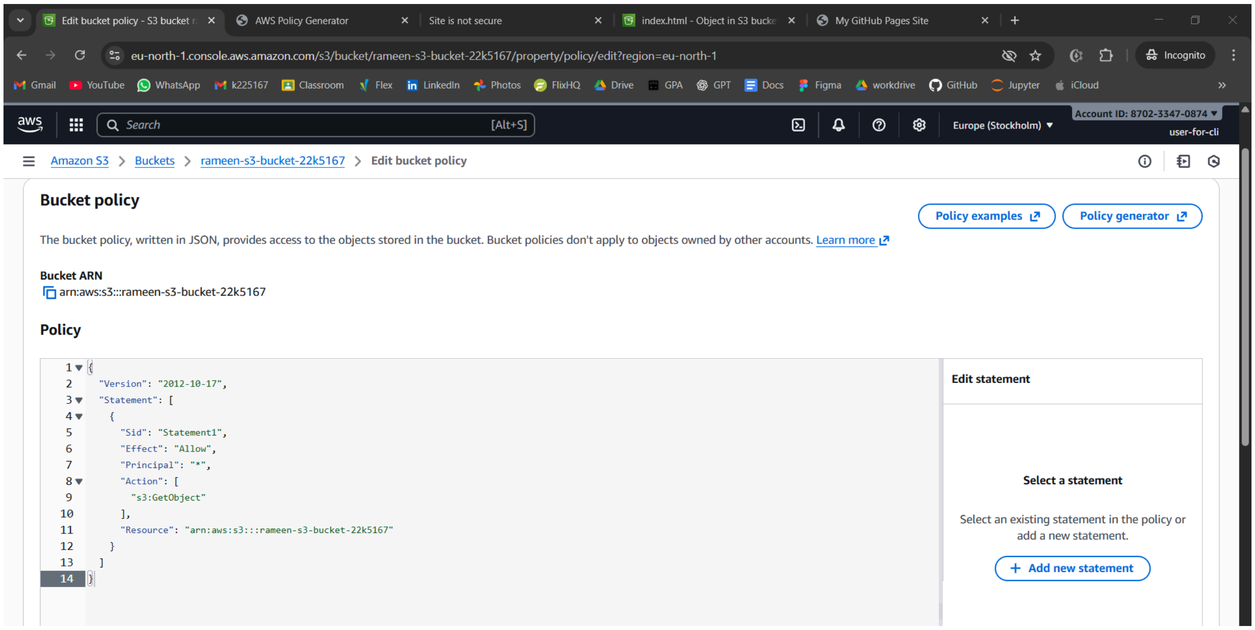


Add statement

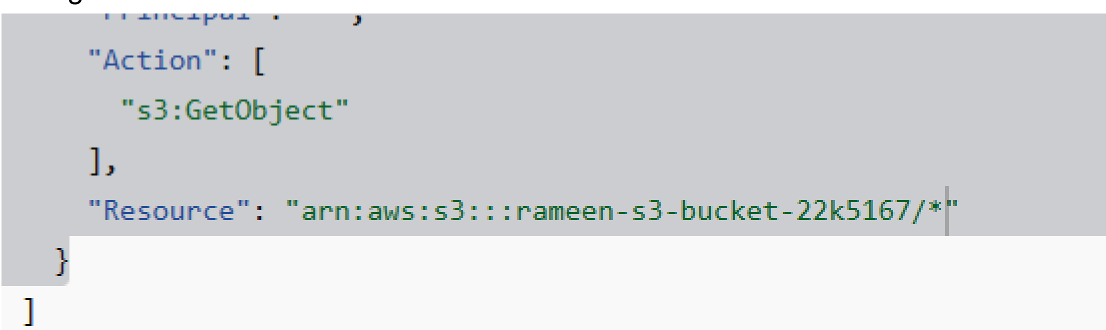



```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "Statement1",
      "Effect": "Allow",
      "Principal": "*",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": "arn:aws:s3:::rameen-s3-bucket-22k5167"
    }
  ]
}
```

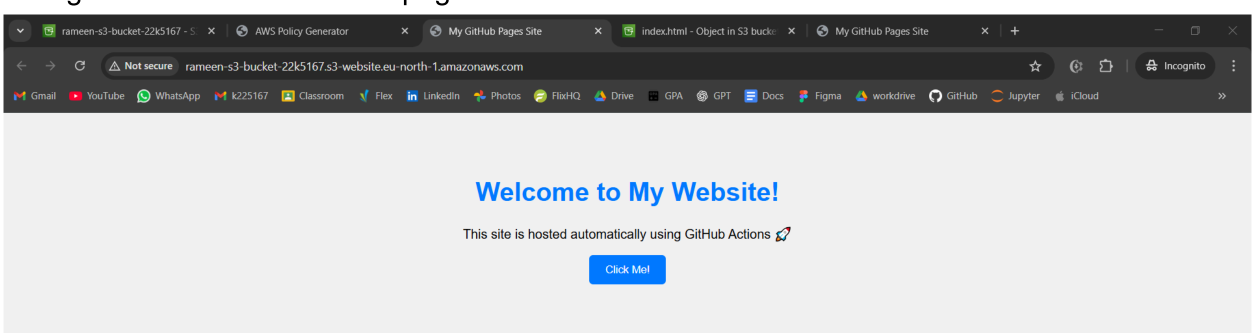
Paste the policy here:



Change this

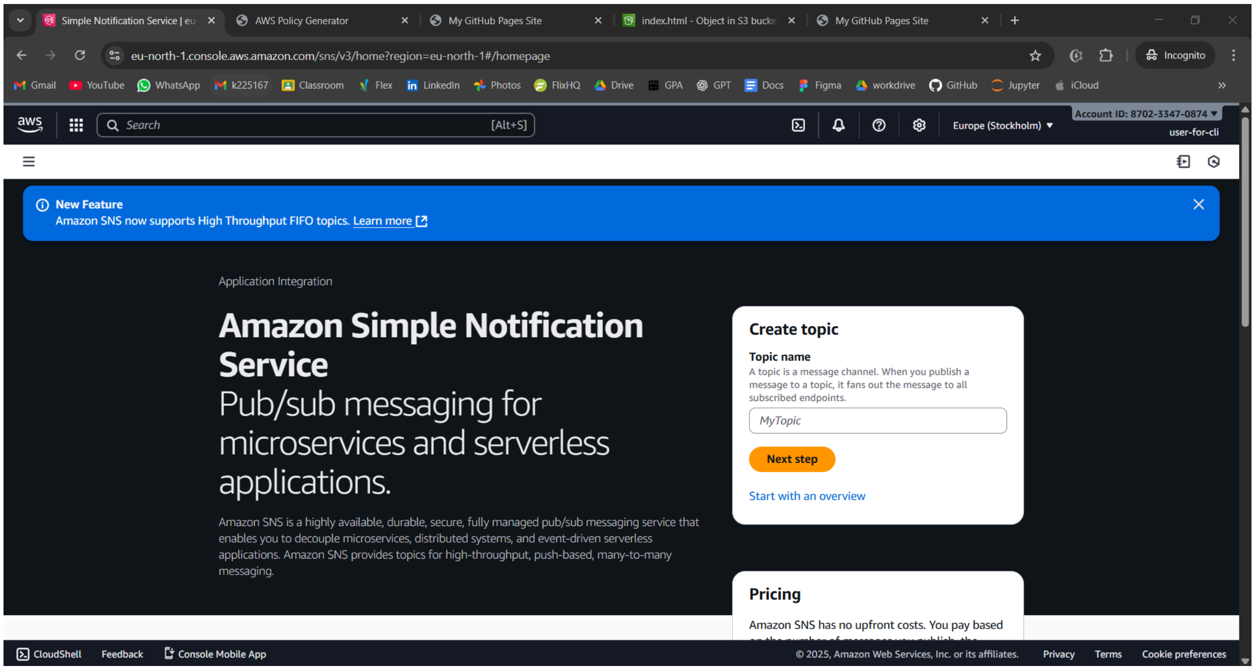


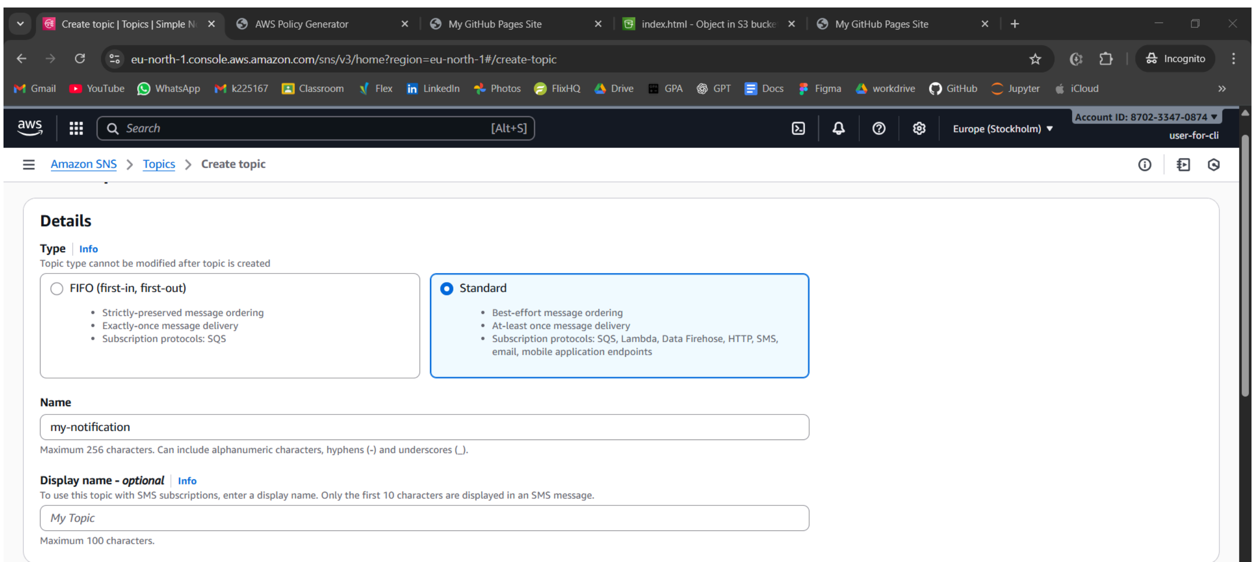
Now go to site and continue to page

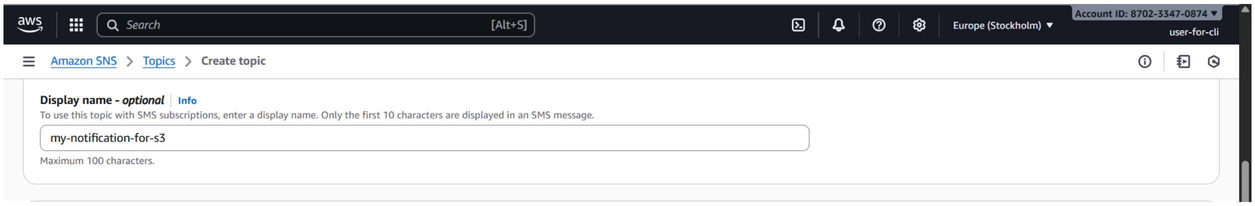


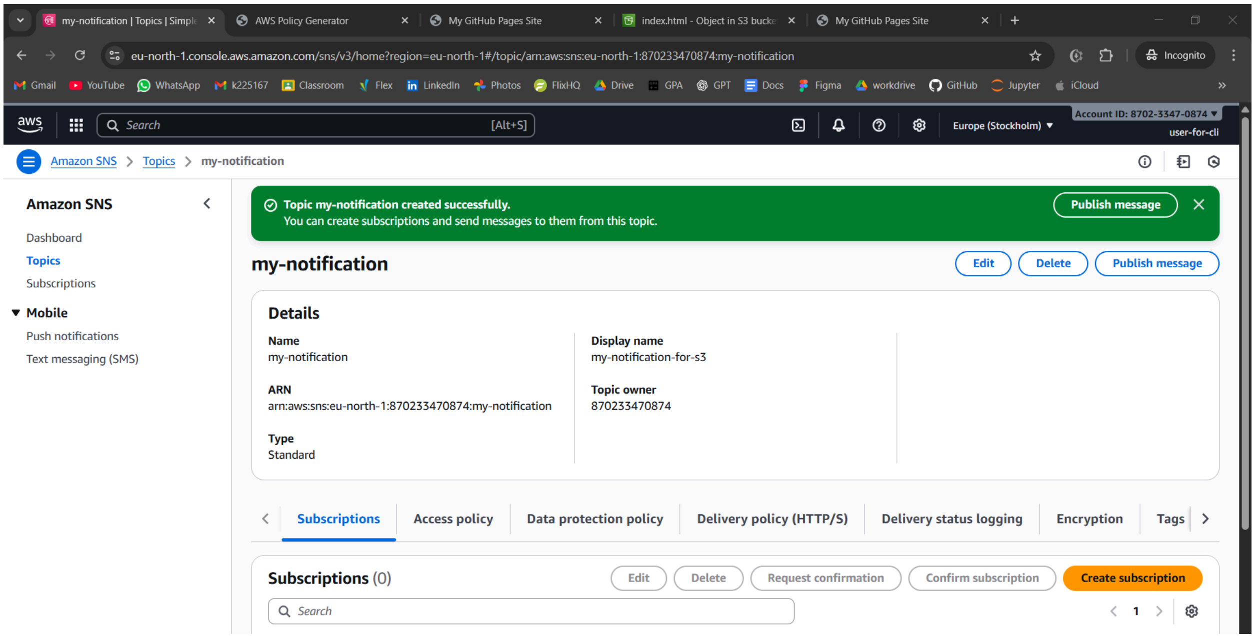
Lambda, sns

Now configuring sns

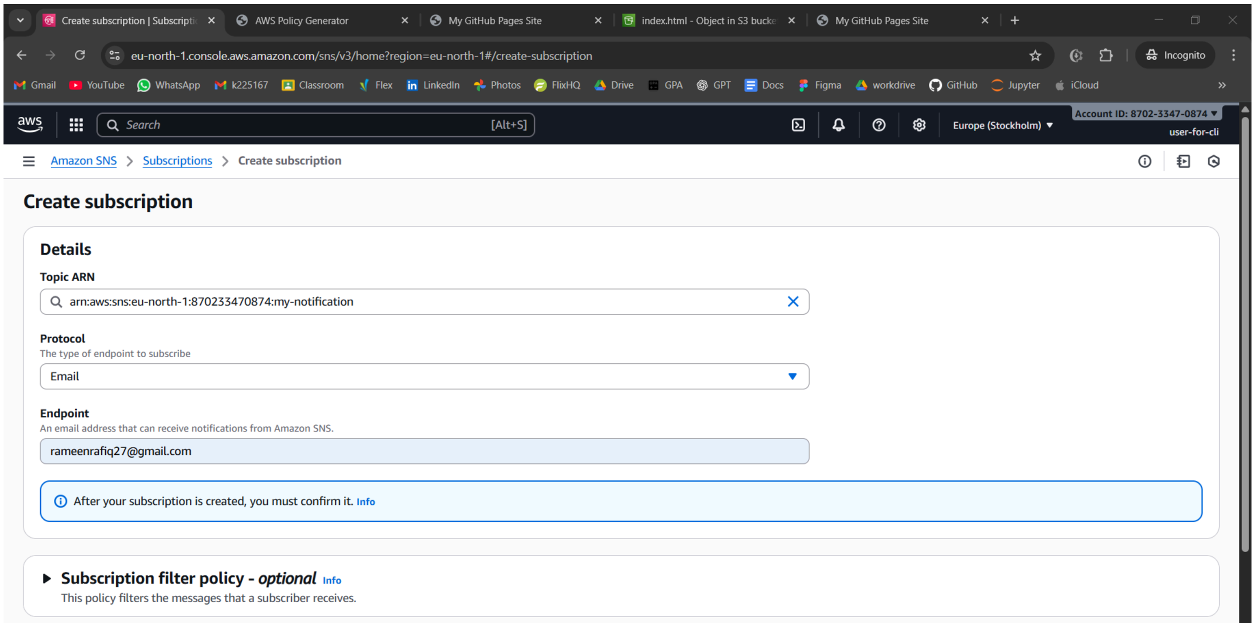

Simple Notification Service | eu
AWS Policy Generator
My GitHub Pages Site
index.html - Object in S3 bucke
My GitHub Pages Site
eu-north-1.console.aws.amazon.com/sns/v3/home?region=eu-north-1#/homepage
Incognito
Gmail YouTube WhatsApp k225167 Classroom Flex LinkedIn Photos FlixHQ Drive GPA GPT Docs Figma workdrive GitHub Jupyter iCloud
Search [Alt+S]
Europe (Stockholm)
Account ID: 8702-3347-0874
user-for-cli
New Feature
Amazon SNS now supports High Throughput FIFO topics. Learn more
Application Integration
Amazon Simple Notification Service
Pub/sub messaging for microservices and serverless applications.
Amazon SNS is a highly available, durable, secure, fully managed pub/sub messaging service that enables you to decouple microservices, distributed systems, and event-driven serverless applications. Amazon SNS provides topics for high-throughput, push-based, many-to-many messaging.
Create topic
Topic name
A topic is a message channel. When you publish a message to a topic, it fans out the message to all subscribed endpoints.
MyTopic
Next step
Start with an overview
Pricing
Amazon SNS has no upfront costs. You pay based
CloudShell
Feedback
Console Mobile App
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

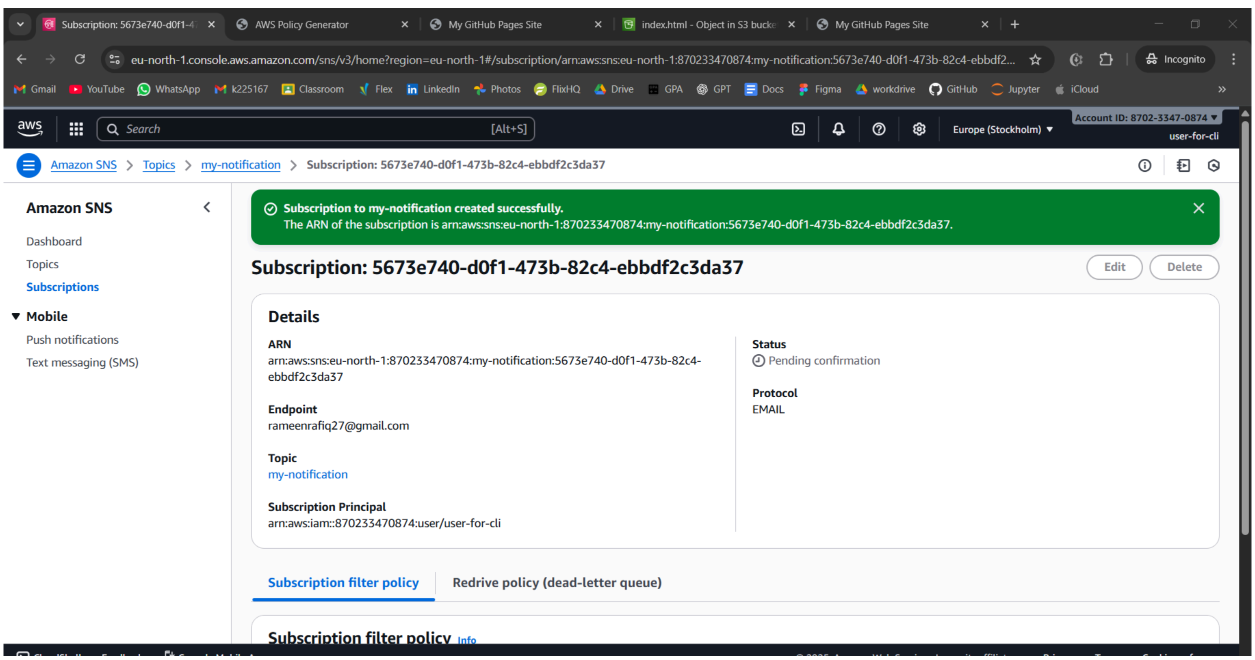

Create topic | Topics | Simple N
AWS Policy Generator
My GitHub Pages Site
index.html - Object in S3 bucke
My GitHub Pages Site
eu-north-1.console.aws.amazon.com/sns/v3/home?region=eu-north-1#/create-topic
Incognito
Gmail YouTube WhatsApp k225167 Classroom Flex LinkedIn Photos FlixHQ Drive GPA GPT Docs Figma workdrive GitHub Jupyter iCloud
Search [Alt+S]
Europe (Stockholm)
Account ID: 8702-3347-0874
user-for-cli
Amazon SNS > Topics > Create topic
Details
Type Info
Topic type cannot be modified after topic is created
FIFO (first-in, first-out)
Strictly-preserved message ordering
Exactly-once message delivery
Subscription protocols: SQS
Standard
Best-effort message ordering
At-least once message delivery
Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints
Name
my-notification
Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).
Display name - optional Info
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.
My Topic
Maximum 100 characters.



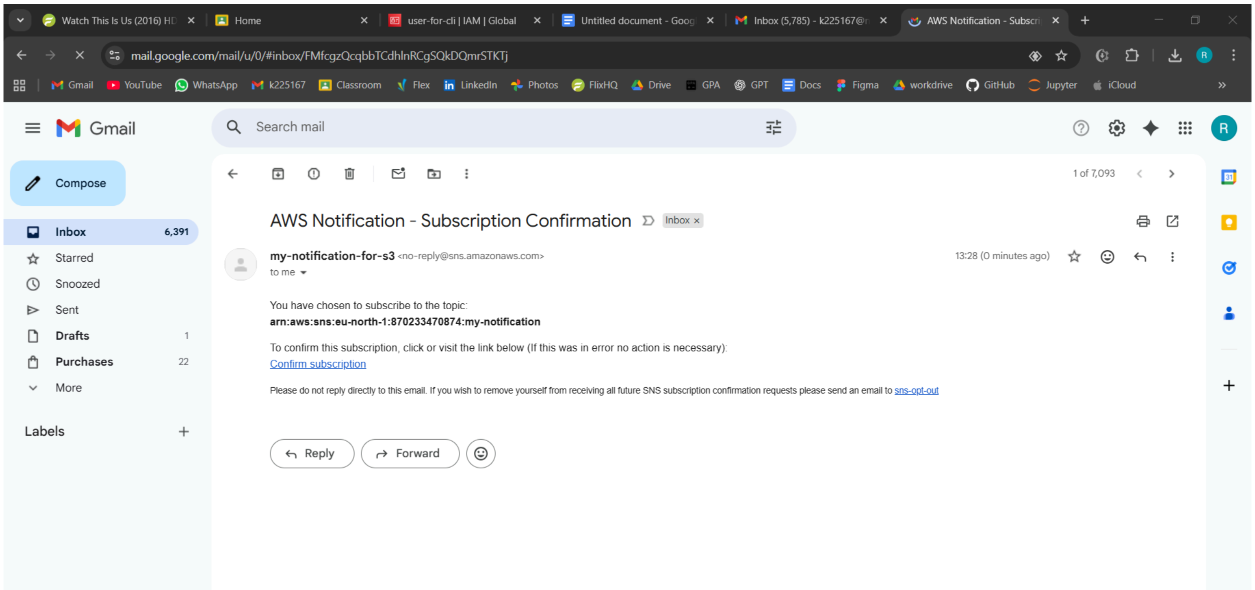


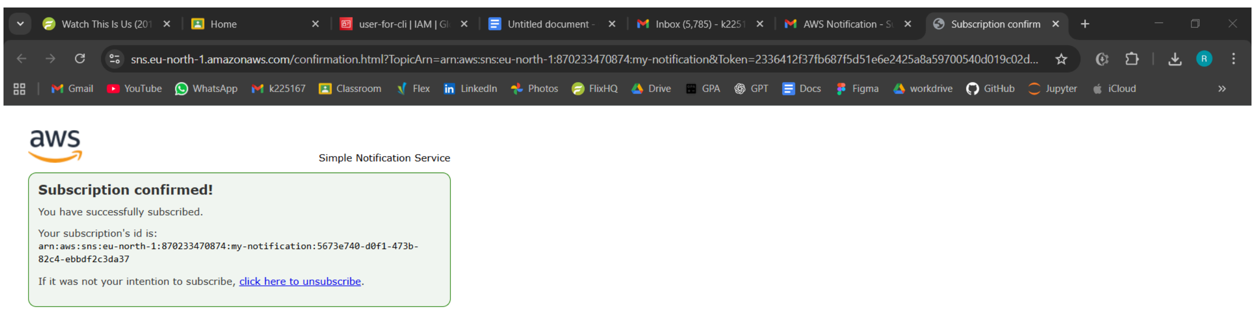
Click on create subscription



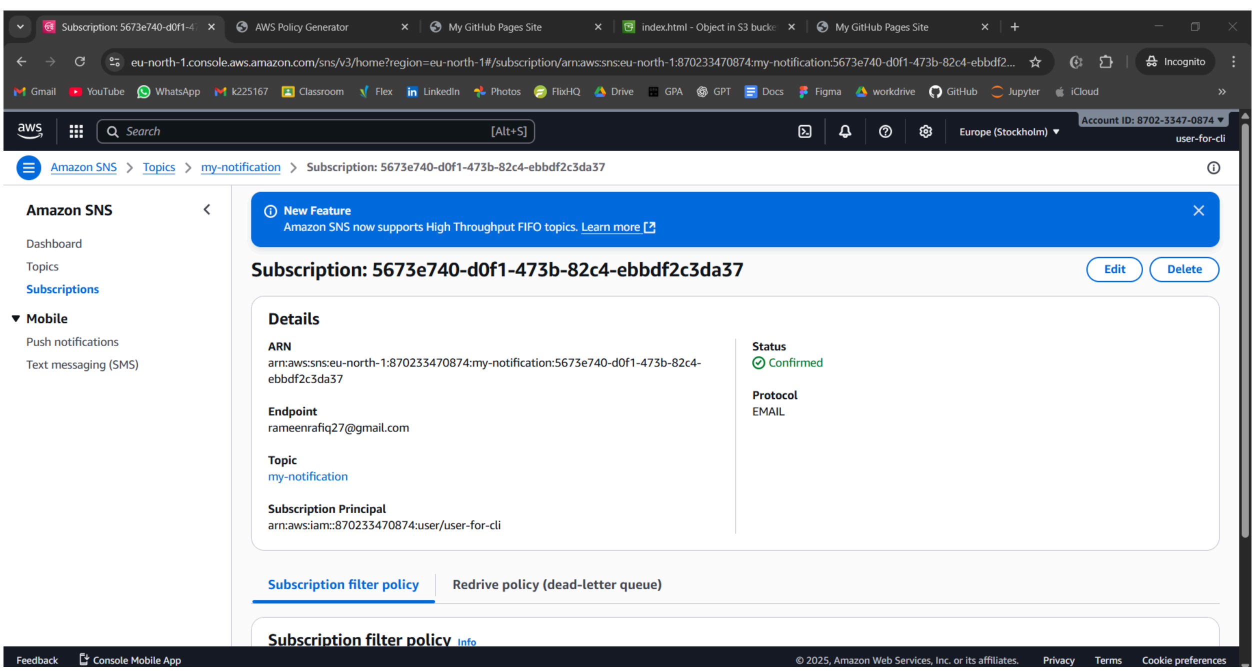


Status is pending

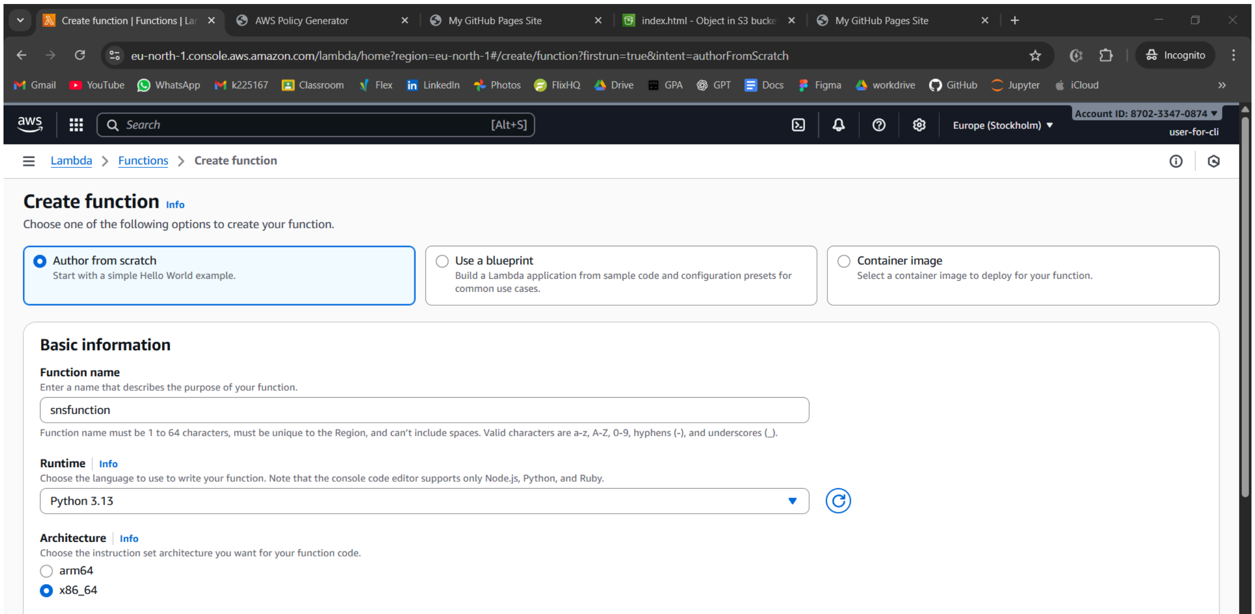


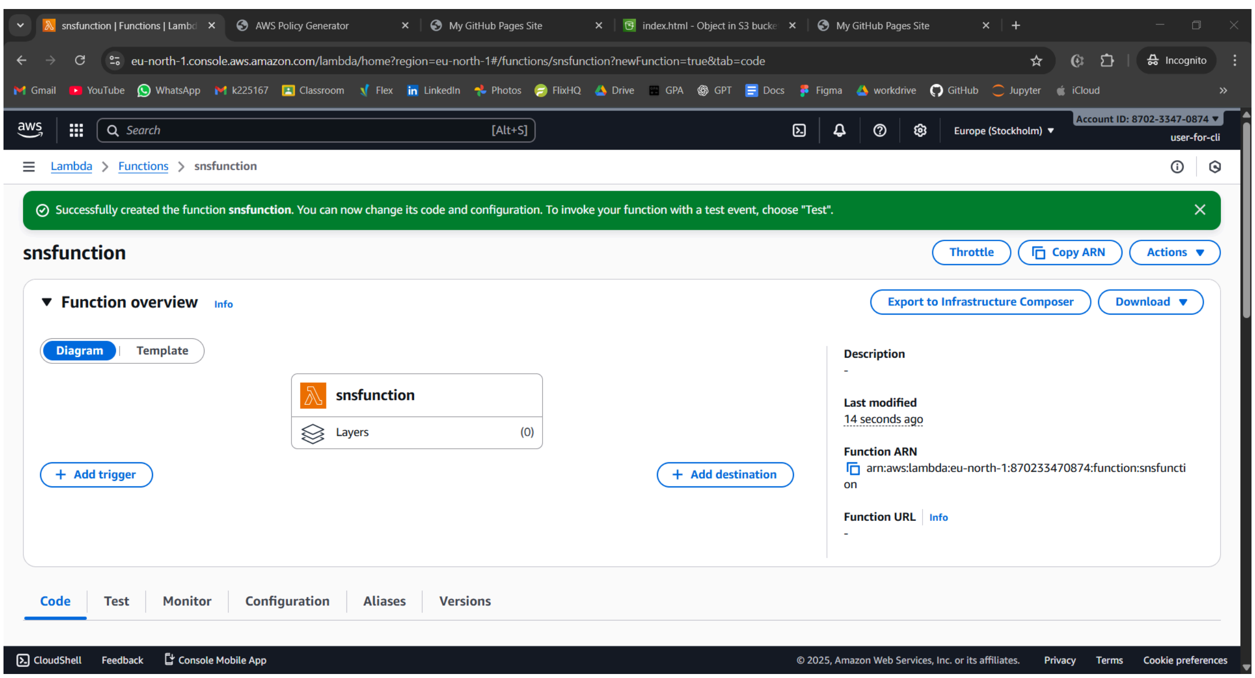


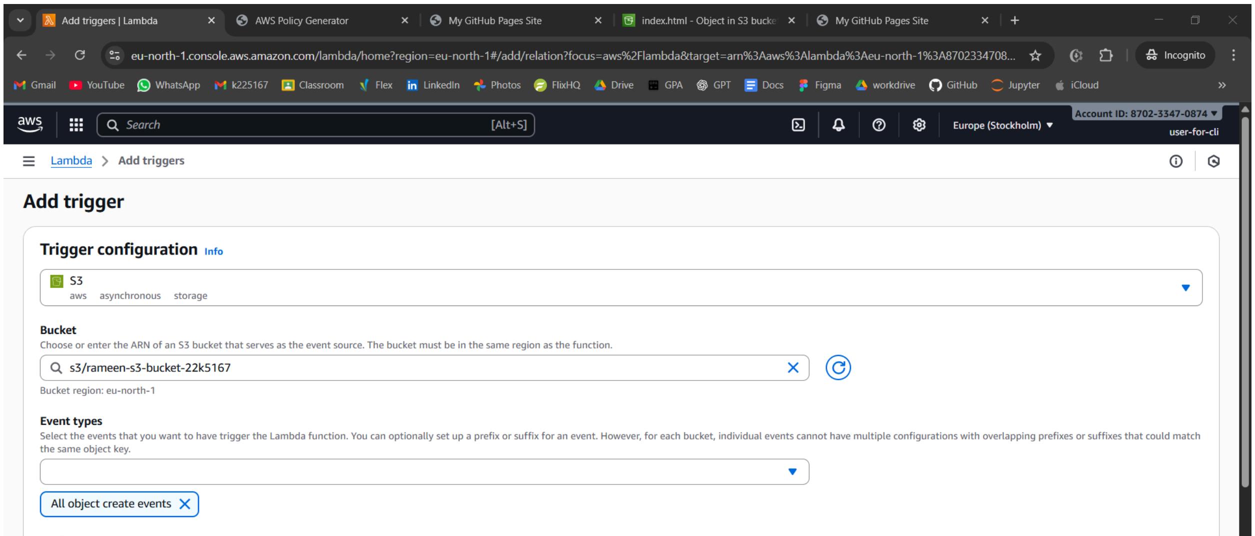
Status is now confirmed

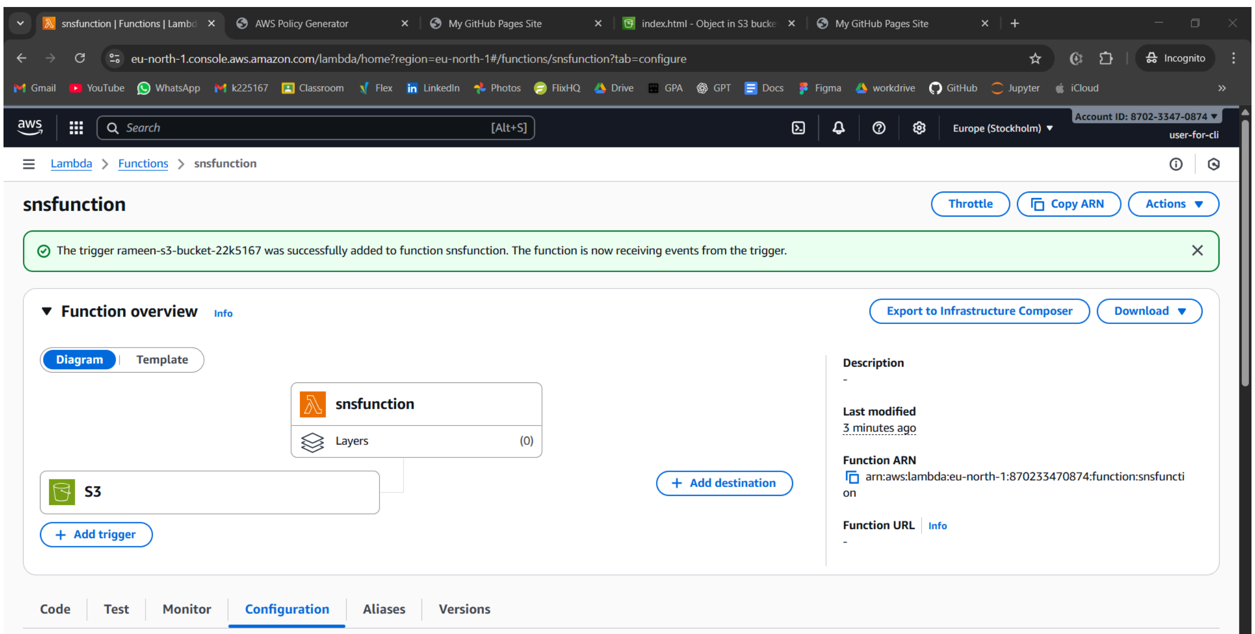


Now we create lambda function and connect them



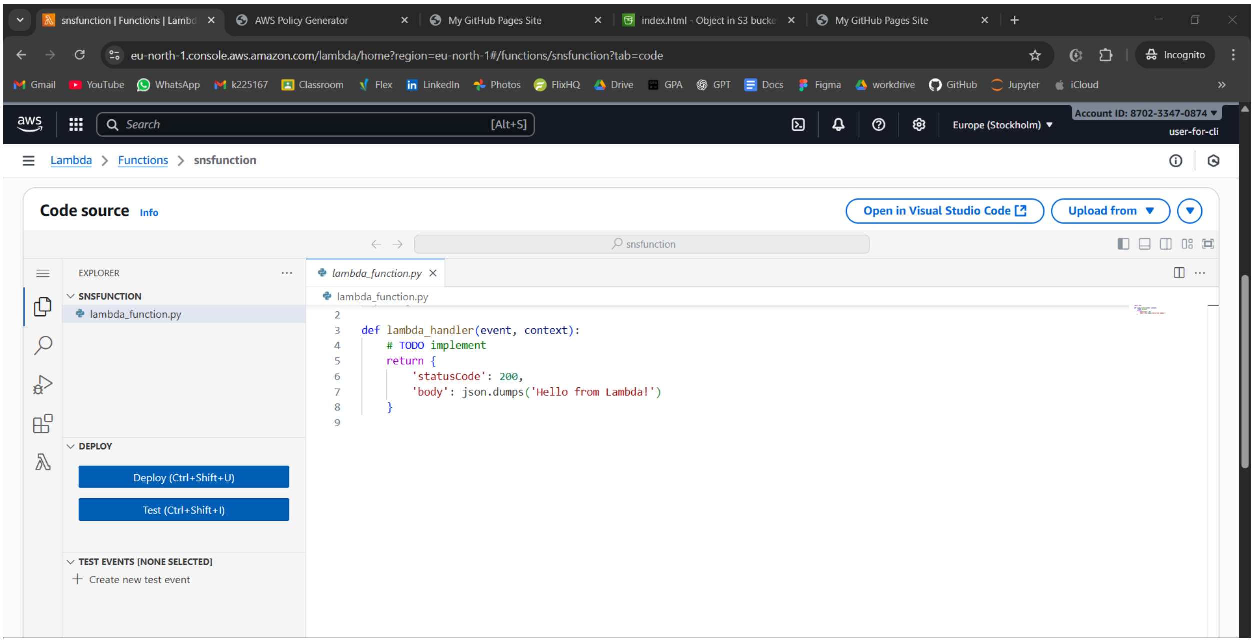




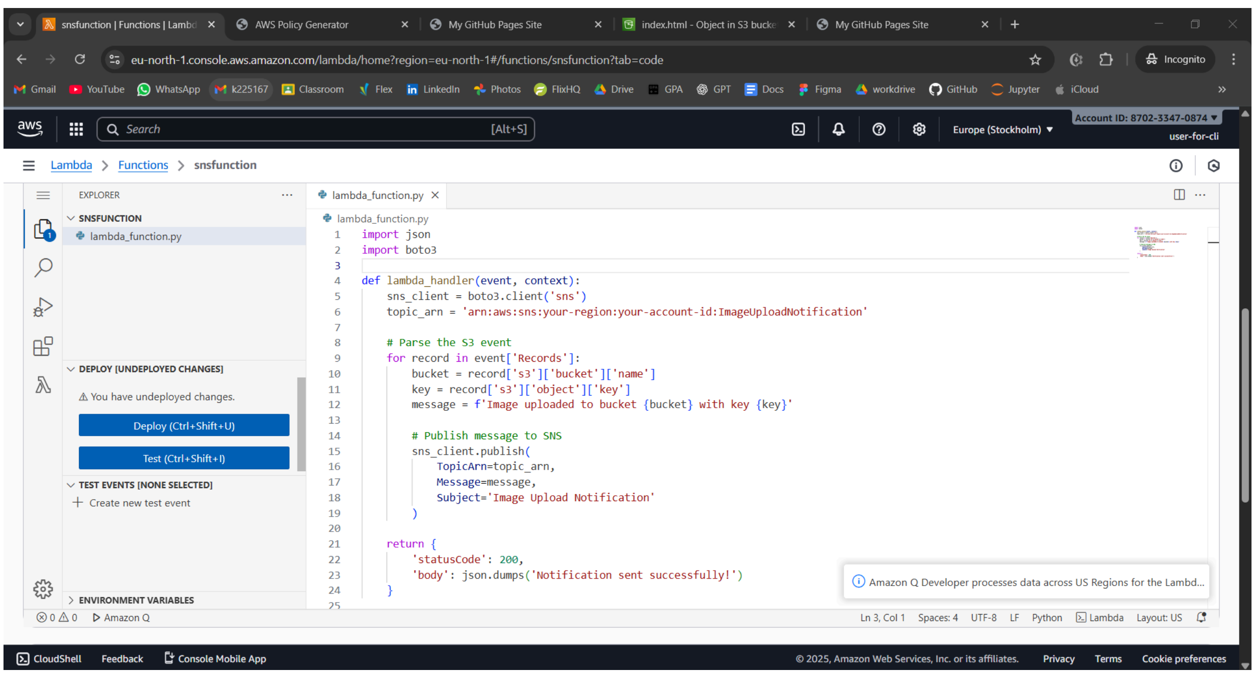


Click on layers

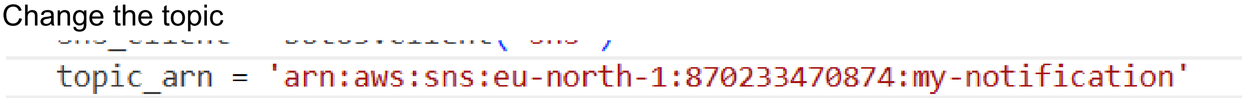
Go to code option



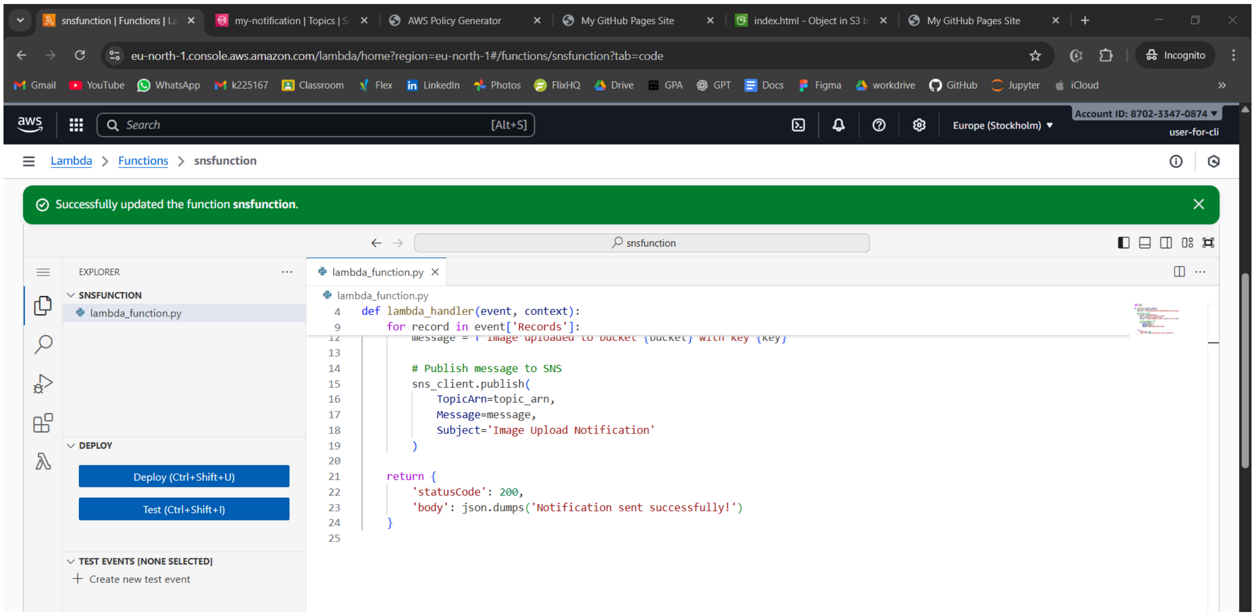
Open the lambda function file in folder
And paste it



Change the topic



We use json bec we are coding in python
Saves as name and prints message
Then deploy

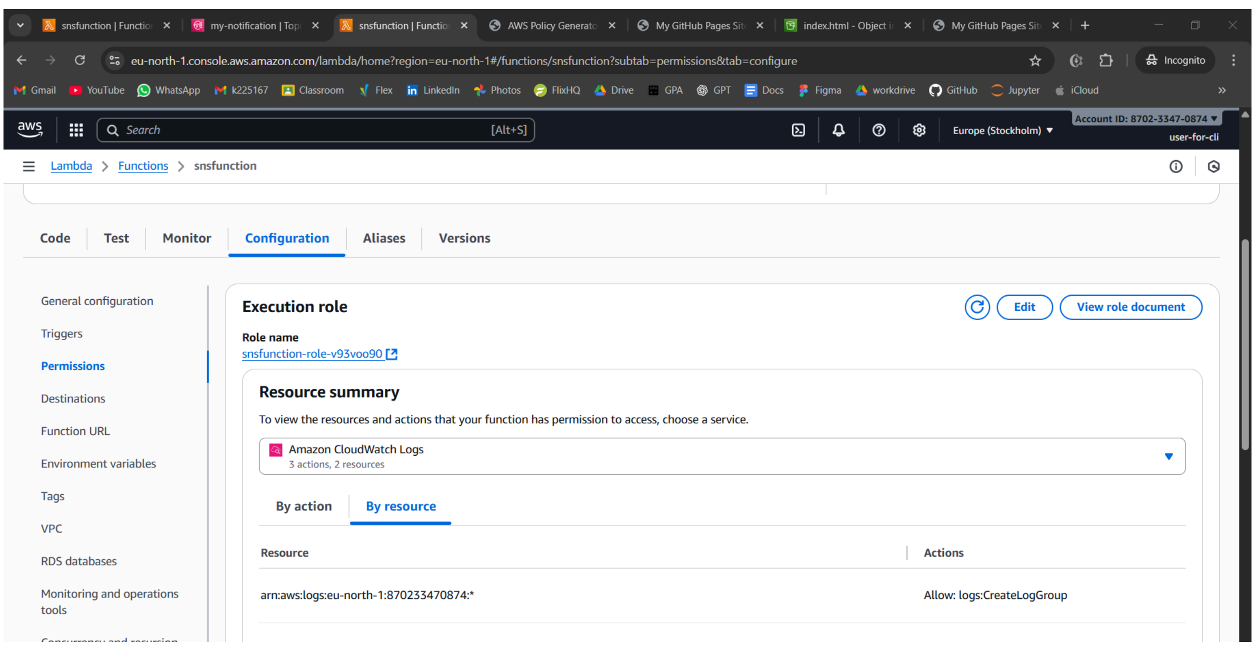


But notification still wont come bec permission not given.

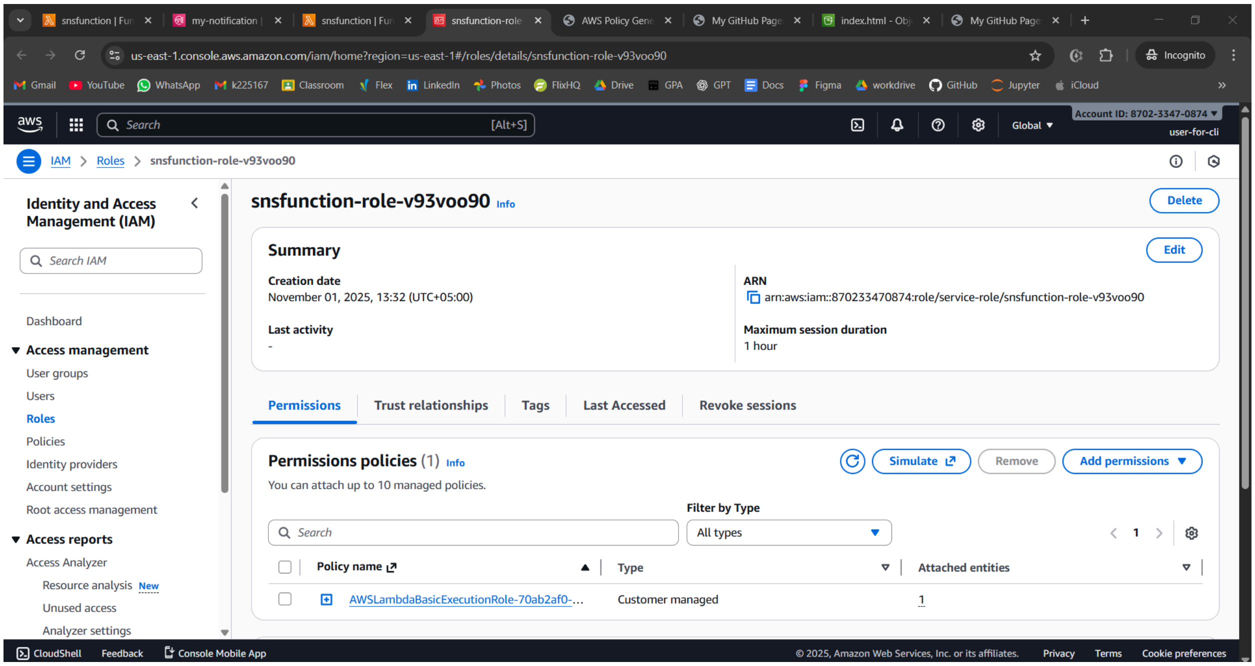
In aws if one service communicates with other, service also needs permission even though it is in same zone. Permissions also for services.

Roles can also be made for services, not only users

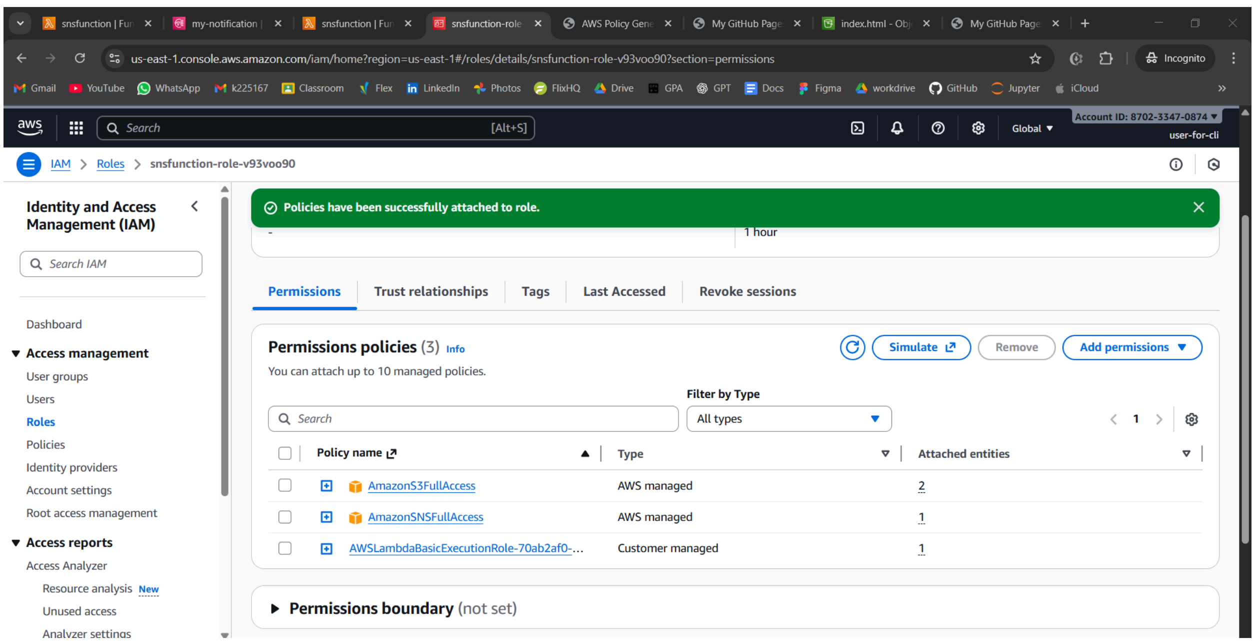
Now we have to give lambda permission to access s3 bucket and sns as well.



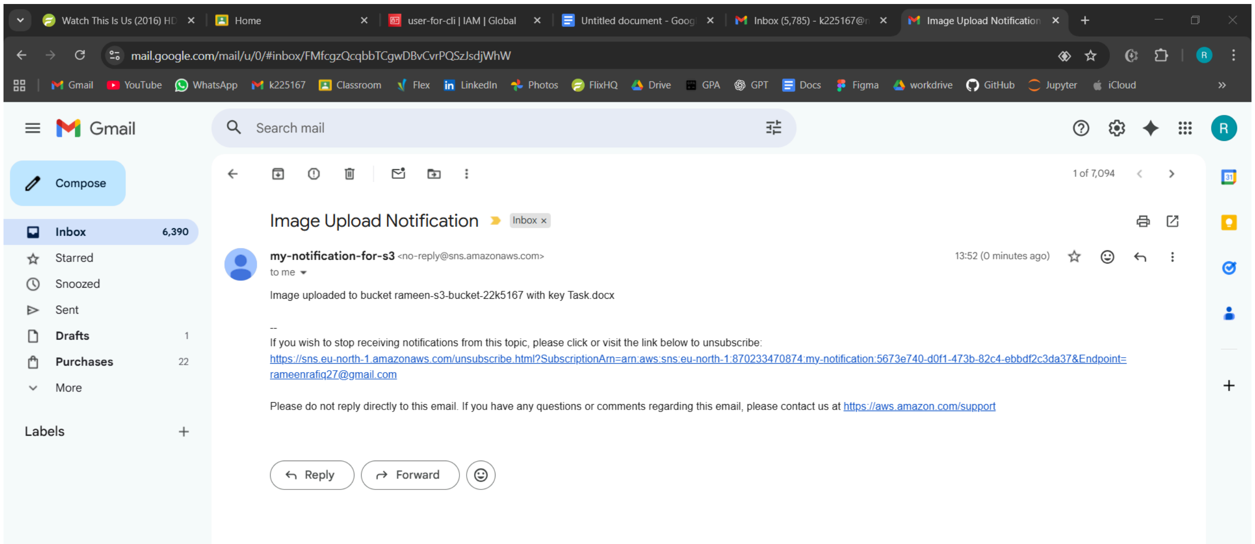
Click on role name



Now we add permission for s3 and sns

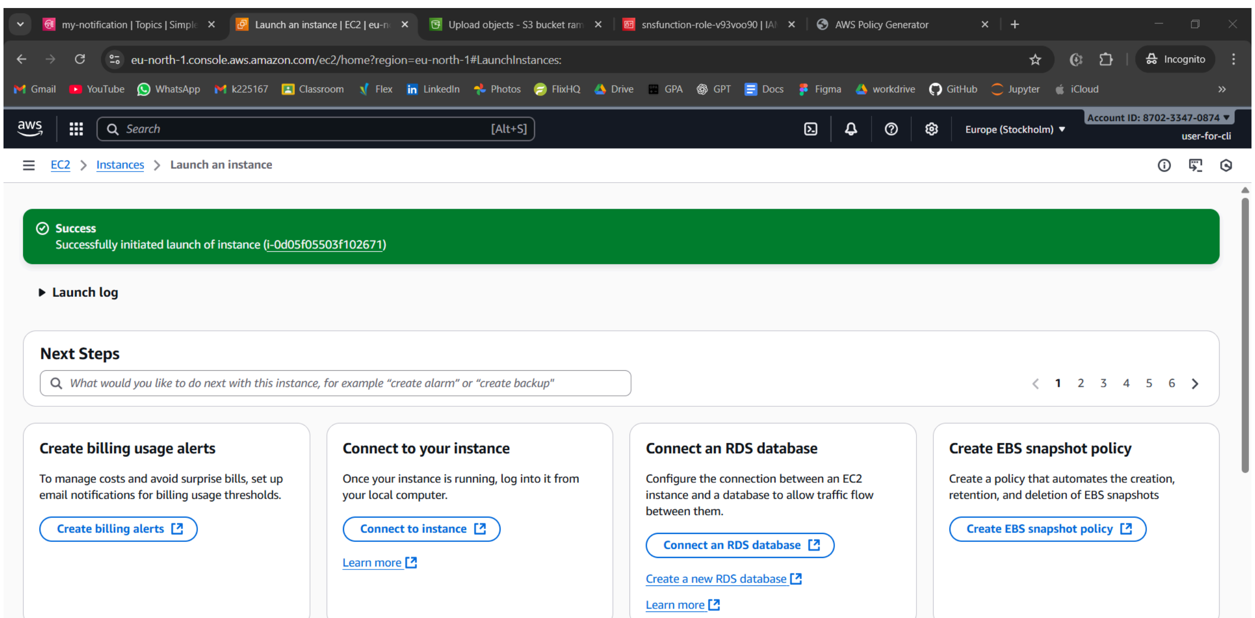


Upload any file to s3 and check email



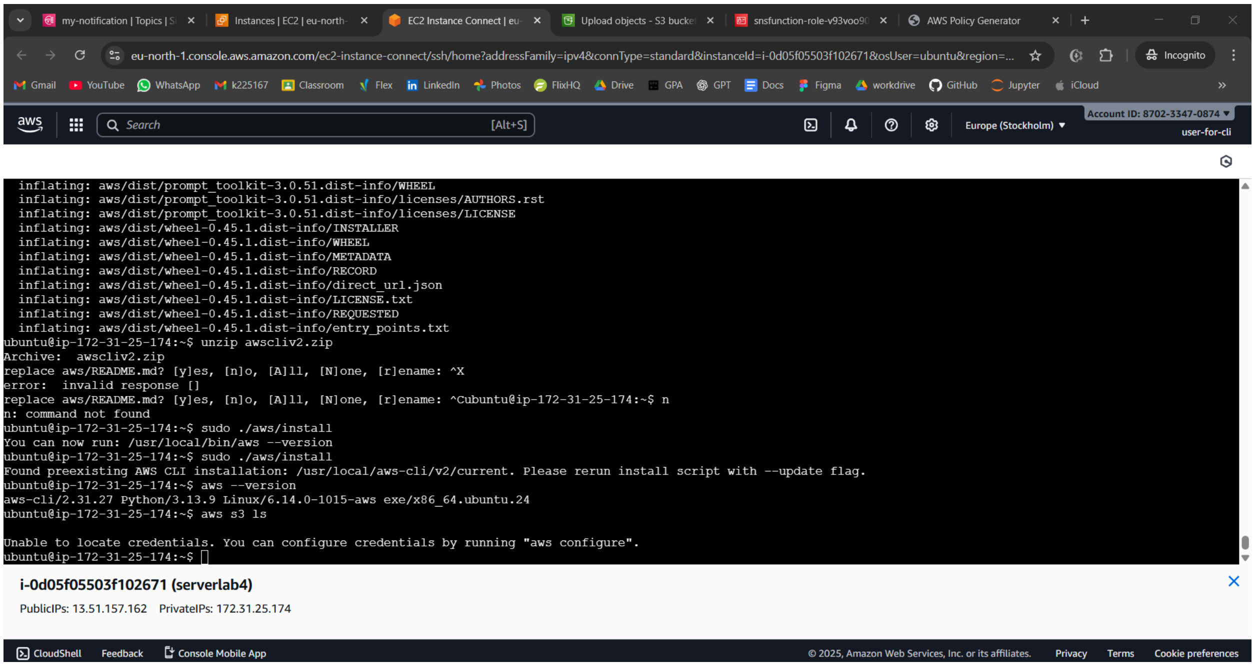
Second actibity

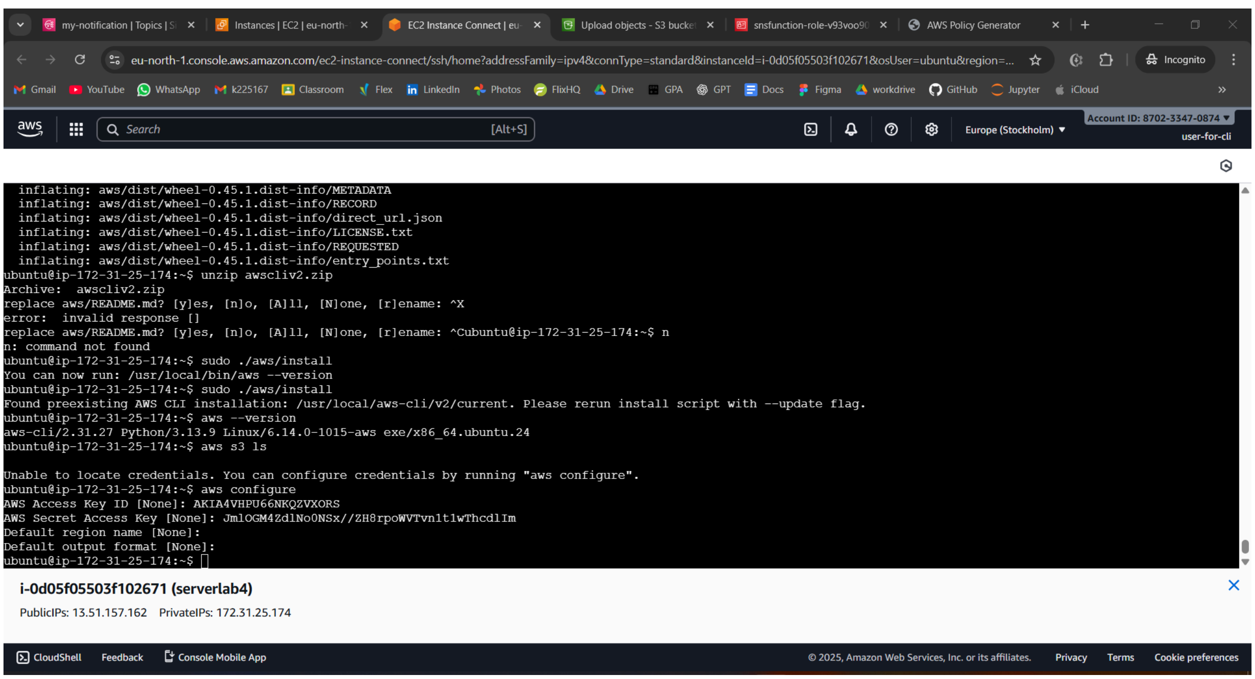
Create ec2



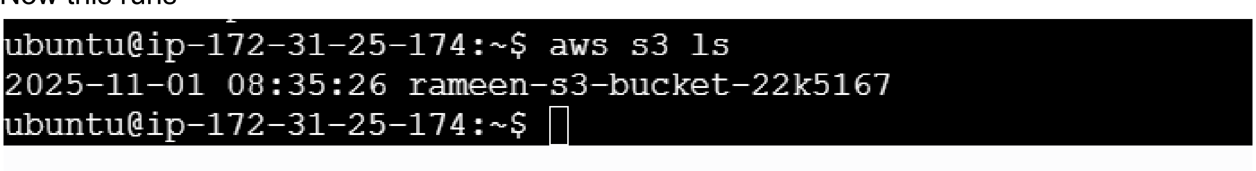
Then connect

Now run all commands

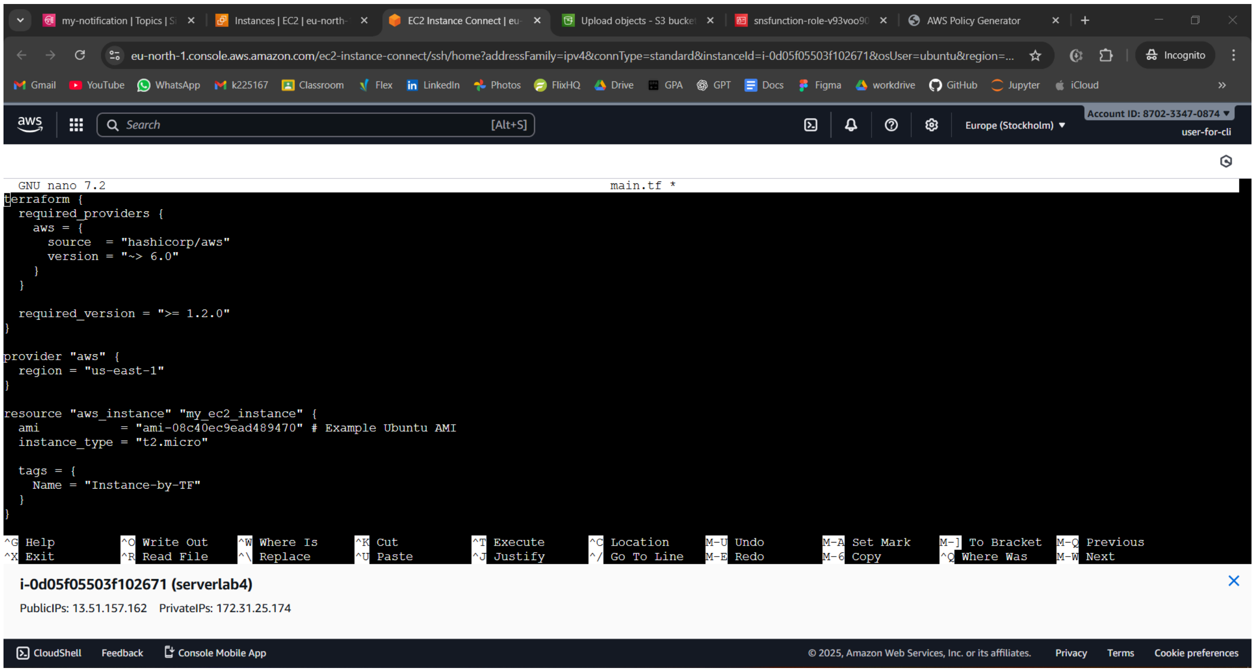




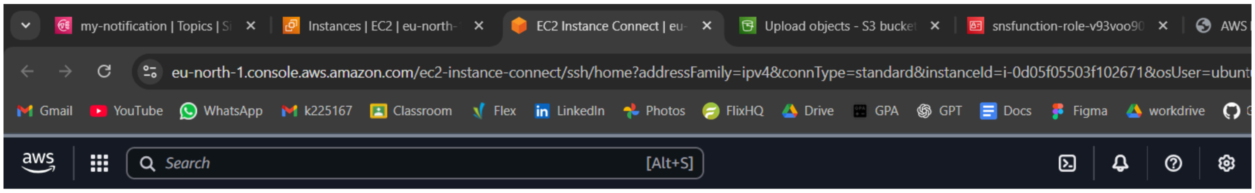
Now this runs



Nano main.tf and paste



Change t2 to t3
Run terraform init
Plan means details of everything to make
Run terraform plan
Known after apply - by default will be selected
Run terraform apply




```
      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + primary_network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.my_ec2_instance: Creating...
aws_instance.my_ec2_instance: Still creating... [00m10s elapsed]
aws_instance.my_ec2_instance: Creation complete after 16s [id=i-015303ef185d0dcb8]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
ubuntu@ip-172-31-25-174:~$
```

i-0d05f05503f102671 (serverlab4)

PublicIPs: 13.51.157.162 PrivateIPs: 172.31.25.174

ec2 instance would have been created now
Change your region to east 1 to view it